Modal Analysis and Stability of Discrete Time and Continuous Time State Space Models

Preliminaries for the Systems Theory Course

Prof. M.Elena Valcher

A.Y. 2024/25

Contents

Chapter 1

Discrete Time State Space Models

1.1 The Tank Example

Hot water and cold water coming from two different sources are conveyed into a water mixer. The water leaving the mixer, with temperature $T_M(\cdot)$, enters a tank. The water temperature in the tank is indicated by $T_S(\cdot)$. We assume that the incoming flow in the tank equals the outgoing one. This ensures that the only effect of the incoming water in the tank is to influence the temperature of the water stored in it, while its level remains unchanged. The figure below illustrates the setting of the overall system.

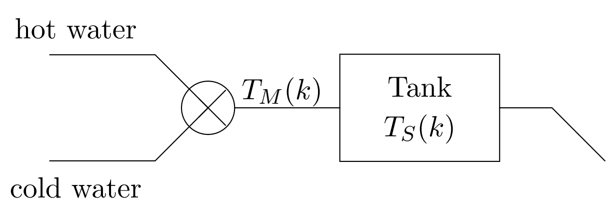


Figure 1.1: *Structure of the Mixer-Tank system.*

Despite of the intrinsically continuous nature of the described phenomenon, we will describe this system with a discrete time model. To do so we assume to measure both the temperature of the outgoing water from the mixer and of the water stored in the tank at equally spaced instants of time of the type kT, with $k \in \mathbb{Z}_+$ and T being a prefixed time interval. For the sake of simplicity, let us denote by $T_S(k)$ and $T_M(k)$ the temperatures $T_S(kT)$ and $T_M(kT)$, respectively.

As a first step, we will focus on the temperature dynamics of the water stored in the tank. It is reasonable to assume for such dynamics, for $k \in \mathbb{Z}_+$, the following model:

$$T_S(k+1) = (1-a)T_S(k) + aT_M(k), \tag{1.1}$$

where a is a positive coefficient, whose value ranges between 0 and 1. In this way, we assume that at the end of the unitary time interval $[k, k+1]$, the temperature of the water in the tank is the weighted average between the temperature of the water in the tank at the starting time k and the temperature of the water coming from the mixer, entering the tank during the time interval $[k, k+1]$. During the interval $[k, k+1]$ the temperature of such water coming from the mixer is supposed to be constant. The value of the coefficient a depends on the intensity of the water flow entering the tank and on the amplitude of the time interval T. Let us suppose that we want the temperature of the water stored in the tank to reach a desired temperature, T_S^*. To meet this goal a control action on the system is needed and the only variable on which a control action can be applied is the temperature of the outgoing water from the mixer. In the following a possible choice for the feedback control law is introduced:

$$T_M(k+1) = T_M(k) - b\ (T_S(k) - T_S^*), \qquad k \in \mathbb{Z}_+, \tag{1.2}$$

where $b \in \mathbb{R}$ is a real, positive coefficient. The idea behind this control law is trivial: the temperature of the outgoing water from the mixer is increased or decreased, with respect to the value of T_M, depending on whether the tank temperature is lower or higher than the desired temperature, respectively. In other words, the difference $T_M(k+1) - T_M(k)$ linearly depends on the deviation $T_S(k) - T_S^*$. By writing the two previous equations (1.1) and (1.2) in matrix form, the following system of (first order) linear difference equations with constant coefficients is obtained:

$$\begin{bmatrix} T_S(k+1) \\ T_M(k+1) \end{bmatrix} = \begin{bmatrix} 1-a & a \\ -b & 1 \end{bmatrix} \begin{bmatrix} T_S(k) \\ T_M(k) \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^*. \tag{1.3}$$

It is worth noticing that a and b are fixed design parameters, depending on the plant, while the desired temperature of the water in the tank T_S^*, may vary according to the operational needs, thus it can be modelled as an input for the overall system. By setting

$$\mathbf{x}(k) := \begin{bmatrix} T_S(k) \\ T_M(k) \end{bmatrix}$$

and $u(k) = T_S^*$, the model in (1.3) can be equivalently rewritten, in a more compact form, as

$$\mathbf{x}(k+1) = F\mathbf{x}(k) + Gu(k). \tag{1.4}$$

The vector $\mathbf{x}(k)$ is called **state vector**, while the vector $u(k)$ (that in this specific case is fixed to a constant value) represents the **input** of the system, namely the variable of our problem that we can freely choose. The system **output**, namely the measured variable, denoted as usual by $y(k)$, is the temperature of the water in the tank, therefore it can be expressed as

$$y(k) = H\mathbf{x}(k), \tag{1.5}$$

where $H = [\,1 \quad 0\,]$. The model described in (1.4)-(1.5) is a **state space model**.

1.2 Discrete Time State Space Models

A generic discrete time, linear and time invariant[1] (LTI) state space model, of **dimension** n, with m **inputs** and p **outputs**, is described by a system of (first order) difference equations of the type

$$\mathbf{x}(k+1) = F\mathbf{x}(k) + G\mathbf{u}(k), \tag{1.6}$$

$$\mathbf{y}(k) = H\mathbf{x}(k) + D\mathbf{u}(k), \tag{1.7}$$

where $\mathbf{x}$ is the n-dimensional state vector, $\mathbf{u}$ is the m-dimensional input vector and $\mathbf{y}$ is the p-dimensional output vector. The matrices $F \in \mathbb{R}^{n\times n}$, $G \in \mathbb{R}^{n\times m}$, $H \in \mathbb{R}^{p\times n}$ and $D \in \mathbb{R}^{p\times m}$ are arbitrary. The variable k is an integer variable representing the time. The first equation (1.6) is called the **(one step) state transition map** and the equation (1.7) is the **output equation**.

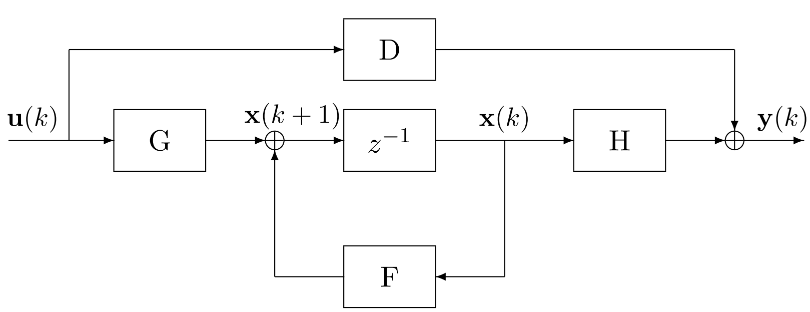
The F matrix is commonly known as **state matrix**. It will be pointed out later how from it, and specifically from its **characteristic polynomial**,

$$\Delta_F(z) \doteq \det(zI_n - F) \tag{1.8}$$

the most significant properties of the state space model (in particular, its internal stability) can be deduced. The zeros of the characteristic polynomial of the matrix F are its **eigenvalues**.

The aforementioned state space model represents a causal LTI system and it is commonly assumed that the system dynamics is observed starting from a certain initial (integer) time instant k_0 (without loss of generality, it is assumed $k_0 = 0$). Moreover, if $D = 0$, i.e. the output $\mathbf{y}(k)$ does not instantaneously depend on the input $\mathbf{u}(k)$, the system is said to be **strictly proper**. The system is denoted for short by the quadruple $\Sigma = (F, G, H, D)$. If it is strictly proper, simply by the triple $\Sigma = (F, G, H)$.

The dynamics of a discrete time state space model can be represented with the following block diagram. In the following, z^{-1} represents the "unit delay system" or "one step delayer" [2].



[1] In the following we will consider only linear and time invariant systems, so this specification will often be omitted.

[2] The reason behind this notation will be clearer later on, once the unit delay will be represented through the $\mathcal{Z}$-transform (see Appendix).

Figure 1.2: *Block diagram of a discrete time state space model.*

1.3 Discrete Time State Space Model Evolution

Let us consider the state space model

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + G\mathbf{u}(k), && (1.9) \\ \mathbf{y}(k) &= H\mathbf{x}(k) + D\mathbf{u}(k), \qquad k \in \mathbb{Z}_+, && (1.10) \end{aligned}$$

where $\mathbf{x}$, $\mathbf{u}$, $\mathbf{y}$ and the matrices F, G, H and D have the aforementioned interpretation. Let us suppose that the initial state $\mathbf{x}(0) = \mathbf{x}_0$ and the control input $\mathbf{u}(k)$, for $k \in \mathbb{Z}_+$, are both known.

In order to the get the expression of the state variable at the generic time instant $k \in \mathbb{Z}_+$, let us explicitly write the state expression for the first time instants $k \in \mathbb{Z}_+$, thus obtaining

$$\begin{aligned} \mathbf{x}(1) &= F\mathbf{x}(0) + G\mathbf{u}(0) = F\mathbf{x}_0 + G\mathbf{u}(0), \\ \mathbf{x}(2) &= F\mathbf{x}(1) + G\mathbf{u}(1) = F^2\mathbf{x}_0 + FG\mathbf{u}(0) + G\mathbf{u}(1), \\ \mathbf{x}(3) &= F\mathbf{x}(2) + G\mathbf{u}(2) = F^3\mathbf{x}_0 + F^2G\mathbf{u}(0) + FG\mathbf{u}(1) + G\mathbf{u}(2), \\ \mathbf{x}(4) &= F\mathbf{x}(3) + G\mathbf{u}(3) = F^4\mathbf{x}_0 + F^3G\mathbf{u}(0) + F^2G\mathbf{u}(1) + FG\mathbf{u}(2) + G\mathbf{u}(3). \end{aligned}$$

The expression of the state variable at the generic time instant $k \in \mathbb{Z}_+$ is given by

$$\mathbf{x}(k) = F^k\mathbf{x}_0 + \sum_{i=0}^{k-1} F^{k-1-i}G\mathbf{u}(i), \qquad k \in \mathbb{Z}_+, \tag{1.11}$$

where, by convention, the summation in (1.11) is assumed to be zero for $k < 1$. According to the above expression (1.11), the output evolution is given by

$$\mathbf{y}(k) = HF^k\mathbf{x}_0 + \sum_{i=0}^{k-1} HF^{k-1-i}G\mathbf{u}(i) + D\mathbf{u}(k), \qquad k \in \mathbb{Z}_+. \tag{1.12}$$

It is immediate to identify the initial state $\mathbf{x}(0) = \mathbf{x}_0$ and the input $\mathbf{u}(k), k \in \mathbb{Z}_+$, as the causes acting on the system, while the state evolution $\mathbf{x}(k)$ and the output evolution $\mathbf{y}(k)$, for $k \in \mathbb{Z}_+$, are the corresponding effects. Since the system is linear, the superposition principle holds, thus both the state and the output evolution can be decomposed in an unforced evolution and a forced one, for $k \geq 0$. This amounts to saying that we can equivalently express $\mathbf{x}(k)$ and $\mathbf{y}(k)$ as

$$\begin{aligned} \mathbf{x}(k) &= \mathbf{x}_\ell(k) + \mathbf{x}_f(k), \\ \mathbf{y}(k) &= \mathbf{y}_\ell(k) + \mathbf{y}_f(k), \quad k \in \mathbb{Z}_+, \end{aligned}$$

where $\mathbf{x}_\ell$ and $\mathbf{y}_\ell$ represent the **unforced state evolution** and the **unforced output evolution**, respectively, while $\mathbf{x}_f$ and $\mathbf{y}_f$ represent the **forced state evolution** and the **forced output evolution**, respectively.

By making a comparison with (1.11) and (1.12) the aforementioned four components can be deduced:

$$\begin{aligned}
\mathbf{x}_\ell(k) &= F^k \mathbf{x}_0,\\
\mathbf{x}_f(k) &= \sum_{i=0}^{k-1} F^{k-1-i} G\mathbf{u}(i),\\
\mathbf{y}_\ell(k) &= HF^k \mathbf{x}_0,\\
\mathbf{y}_f(k) &= \sum_{i=0}^{k-1} HF^{k-1-i} G\mathbf{u}(i) + D\mathbf{u}(k).
\end{aligned}$$

We want now to give a mathematical interpretation to the last equality, namely the one corresponding to the forced output evolution. To meet this goal, let us consider as a first step, the case in which the system has just one input, i.e. $m = 1$, and let us consider the sequence

$$W(k) \doteq \begin{cases} 0, & \text{for } k \in \mathbb{Z}, k < 0;\\ D, & \text{for } k = 0;\\ HF^{k-1}G, & \text{for } k \in \mathbb{Z}, k \geq 1. \end{cases}$$

By introducing the **Kronecker delta** (or discrete unit impulse)

$$\delta(k) \doteq \begin{cases} 1, & \text{for } k = 0;\\ 0, & \text{for } k \in \mathbb{Z}, k \neq 0; \end{cases}$$

and the **discrete unit step**

$$\delta_{-1}(k) \doteq \begin{cases} 1, & \text{for } k \in \mathbb{Z}, k \geq 0;\\ 0, & \text{for } k \in \mathbb{Z}, k < 0; \end{cases}$$

the sequence $W(k)$ can be rewritten as follows:

$$W(k) = D\delta(k) + HF^{k-1}G\ \delta_{-1}(k-1).$$

If we now consider the expression of the forced evolution of the system output when the input signal $u(k) = \delta(k)$ is applied, it is immediate to notice that it is zero for $k < 0$ and that

$$y_f(k) = \begin{cases} D, & \text{for } k = 0;\\ HF^{k-1}G, & \text{for } k \in \mathbb{Z}_+, k \geq 1. \end{cases}$$

This is the reason why, the sequence $W(k)$ represents the **impulse response** of the system. Let us consider now the case when m is a generic positive integer and let us suppose to apply to the system the input sequence

$$\mathbf{u}^{(i)}(k) = \mathbf{e}_i\ \delta(k),$$

where $\mathbf{e}_i$ is the i-th canonical vector $\mathbf{e}_i \in \mathbb{R}^m$ (namely the vector whose components are all equal to zero except for the i-th one that is equal to 1). In this case the forced output evolution of the system is given by

$$W_i(k) \doteq \mathbf{y}^{(i)}(k) = D\mathbf{e}_i\ \delta(k) + HF^{k-1}G\mathbf{e}_i\ \delta_{-1}(k-1).$$

Since this mathematical relation holds for every $i \in \{1, 2, \dots, m\}$, the matrix sequence (taking values in $\mathbb{R}^{p\times m}$)

$$W(k) \doteq D\delta(k) + HF^{k-1}G\ \delta_{-1}(k-1),$$

with $W_i(k)$ the i-th column of $W(k)$, is called the **impulse response** of the system and the aforementioned forced output evolution, can be interpreted as the **discrete convolution** between the input sequence and the impulse response of the system, i.e.

$$\mathbf{y}_f(k) = \sum_{i=0}^{k} W(k-i)\mathbf{u}(i) = [W * \mathbf{u}](k), \qquad k \in \mathbb{Z}_+.$$

The relation above follows from the causality of the system and from the assumption that the input sequence is zero for $k \in \mathbb{Z}_-$. It is worth noticing that this hypothesis is necessary in order to not account for the contribution of the input samplings $\mathbf{u}(k)$ for $k < 0$ in the system forced evolution, since their effect is already taken into account in the initial state $\mathbf{x}(0)$ appearing in the unforced evolution.

To conclude notice that, when a generic integer is assumed as initial time instant, the state and the output expression at the generic time instant $k \geq k_0$, are given by:

$$\begin{aligned} \mathbf{x}(k) &= F^{k-k_0}\mathbf{x}(k_0) + \sum_{i=k_0}^{k-1} F^{k-1-i}G\mathbf{u}(i), \\ \mathbf{y}(k) &= HF^{k-k_0}\mathbf{x}(k_0) + \sum_{i=k_0}^{k-1} HF^{k-1-i}G\mathbf{u}(i) + D\mathbf{u}(k). \end{aligned}$$

1.4 Jordan Form of a Matrix and Unforced Evolution

As already shown in the previous section, the expressions of the unforced state evolution and unforced output evolution components, starting from the initial condition $\mathbf{x}(0) = \mathbf{x}_0$, are given by

$$\begin{aligned} \mathbf{x}_\ell(k) &= F^k\mathbf{x}_0, \\ \mathbf{y}_\ell(k) &= HF^k\mathbf{x}_0, \qquad k \in \mathbb{Z}_+. \end{aligned}$$

We wonder now, how to calculate the generic k-th power of a given matrix F (this calculation is also useful for the evaluation of the state and output forced evolution components). The examples reported in the following show that the calculation of the power F^k for a generic $k \in \mathbb{Z}_+$ is not trivial, in general.

Example 1.4.1 Let

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

Since F is a diagonal matrix, the expression of its generic k-th power is immediate to derive:

$$F^k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (-1)^k & 0 \\ 0 & 0 & 2^k \end{bmatrix}, \qquad \forall\, k \in \mathbb{Z}_+.$$

♠

Example 1.4.2 Let

$$F = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

In this case the generic k-th power of the matrix F is slightly more difficult to be computed with respect to the previous example. However, by exploiting the block-diagonal structure of the matrix, we notice that the calculation of the k-th power of the matrix F can be easily derived starting from the calculation of the k-th power of its diagonal block

$$F_1 \doteq \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

By noticing that

$$F_1^2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad F_1^3 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}, \quad F_1^4 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix},$$

it is immediate to deduce that

$$F_1^k = \begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix}, \qquad \forall\, k \in \mathbb{Z}_+.$$

Consequently,

$$F^k = \begin{bmatrix} 1 & -k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3^k \end{bmatrix}, \qquad \forall\, k \in \mathbb{Z}_+.$$

♠

Example 1.4.3 Let

$$F = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} \in \mathbb{R}^{2\times 2}.$$

In this case, the evaluation of the first powers of the matrix F is not so helpful for the calculation of the generic expression of F^k. However, the calculation of the characteristic polynomial of F, namely $\Delta_F(z) = \det(zI_2 - F) = z^2 - 5z + 6 = (z-2)(z-3)$, shows that the matrix F has two distinct real eigenvalues, and thus the matrix is **diagonalizable**, by this meaning that F can be reduced to a diagonal form by means of a **similarity transformation**. It is well known that the columns of any matrix T representing the similarity transformation that transforms F into a diagonal matrix are the coordinates of a family of eigenvectors corresponding to the eigenvalues appearing on the diagonal of the transformed matrix (taken in the appropriate order). Therefore, we can assume as (non singular) transformation matrix T, the matrix

$$T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

whose first column is an eigenvector of F associated with the eigenvalue $\lambda_1 = 2$ and whose second column is an eigenvector of F related to $\lambda_2 = 3$. It follows that

$$J \doteq T^{-1}FT = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Therefore, from $F = TJT^{-1}$ it immediately follows that

$$F^k = TJ^kT^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}^k \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2^{k+1} - 3^k & 2\cdot 3^k - 2^{k+1} \\ 2^k - 3^k & 2\cdot 3^k - 2^k \end{bmatrix}.$$

♠

To sum up, in order to evaluate the generic power of a given matrix F we have exploited, in the last example, a similarity transformation that allowed us to transform the matrix F into a diagonal matrix for which the evaluation of the generic k-th power is much simpler. Unfortunately, not all matrices are diagonalizable. However there exists a matrix canonical form, generalization of the diagonal form, to which every square matrix (both with real and complex coefficients) can be reduced via a similarity transformation. For such a canonical form, the computation of the generic k-th power is still manageable. Such a canonical form is known in the literature as **Jordan form**.

Definition 1.4.4 *A matrix $F \in \mathbb{C}^{n\times n}$ is* **in Jordan form** *if it is a block diagonal matrix*

$$F = \begin{bmatrix} \hat{F}_1 & & & \\ & \hat{F}_2 & & \\ & & \ddots & \\ & & & \hat{F}_s \end{bmatrix},$$

in which each diagonal block $\hat{F}_j$ is a **Jordan miniblock***, by this meaning a square* *matrix of the type*

$$\hat{F}_j = J_{\hat{\lambda}_j} = \begin{bmatrix} \hat{\lambda}_j & 1 & & & \\ & \hat{\lambda}_j & 1 & & \\ & & \ddots & \ddots & \\ & & & \hat{\lambda}_j & 1 \\ & & & & \hat{\lambda}_j \end{bmatrix},$$

for some $\hat{\lambda}_j \in \mathbb{C}$.

Notice that the coefficients $\hat{\lambda}_j$ appearing on the diagonal of the Jordan miniblocks are the eigenvalues of the matrix F, as it can be easily verified from the evaluation of its characteristic polynomial. Moreover, if $\lambda_1, \lambda_2, \ldots, \lambda_r$ are the distinct eigenvalues of F, it is immediate to notice that $s \geq r$.

Finally, it is worth noticing that every diagonal matrix is in Jordan form: the diagonal elements, in fact, can be thought of as Jordan miniblocks of dimension one associated with the eigenvalues of the matrix.

Generally, it is common practice to arrange the miniblocks of a matrix in Jordan form (by means of a permutation transformation) in such a way that the miniblocks associated with the same eigenvalues are consecutive and arranged in descending order of size. In other words, a matrix expressed in Jordan form, with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_r$, is usually written as follows:

$$F = \begin{bmatrix} F_1 & & & \\ & F_2 & & \\ & & \ddots & \\ & & & F_r \end{bmatrix},$$

where

$$F_i = \begin{bmatrix} J_{i,1} & & & \\ & J_{i,2} & & \\ & & \ddots & \\ & & & J_{i,s_i} \end{bmatrix}$$

and $J_{i,\ell}$ is the ℓ-th Jordan miniblock associated with the i-th eigenvalue, λ_i, of dimension $n_{i,\ell}$, namely

$$J_{i,\ell} = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_i & 1 \\ & & & & \lambda_i \end{bmatrix} \in \mathbb{C}^{n_{i,\ell} \times n_{i,\ell}}.$$

Moreover it holds that

$$n_{i,1} \geq n_{i,2} \geq \ldots \geq n_{i,s_i}.$$

In the following we introduce some information related to the Jordan form in order to clarify the importance of its role in the evaluation of the generic power of a square matrix F, as well as the numerical evaluation of the state and output vectors during their unforced evolution at the generic time instant k.

Let $\lambda_1, \lambda_2, \dots, \lambda_r$ be the distinct eigenvalues of $F \in \mathbb{C}^{n\times n}$. Two important concepts are associated with them: the **algebraic multiplicity** and the **geometric multiplicity**. The algebraic multiplicity of an eigenvalue λ_i of the matrix F is its multiplicity as zero of the characteristic polynomial of F, $\Delta_F(z) = \det(zI_n - F)$. The geometric multiplicity of λ_i, instead, represents the dimension of the eigenspace $\ker(\lambda_i I_n - F)$, associated with the eigenvalue λ_i.

When F is written in Jordan form, by assuming, for the sake of simplicity, that its miniblocks are ordered according to the aforementioned criterion, it is immediate to deduce the geometric and algebraic multiplicity of its eigenvalues:

- the algebraic multiplicity of λ_i corresponds to the dimension of the block F_i that includes all the Jordan miniblocks associated with λ_i and thus to $\sum_{\ell=1}^{s_i} n_{i,\ell}$;
- the geometric multiplicity of λ_i is, by definition, the dimension of $\ker(\lambda_i I_n - F)$ and thus corresponds to $n - \mathrm{rank}(\lambda_i I_n - F)$, where $\mathrm{rank}(M)$ represents the rank of the matrix M. On the other hand the rank of the matrix $\lambda_i I_n - F$, coincides, due to the particular structure of F, with the number of the nonzero columns in $\lambda_i I_n - F$ and thus with (n - number of Jordan miniblocks associated with λ_i) $= n - s_i$. In the light of these considerations, it is possible to identify the geometric multiplicity of λ_i with the number of Jordan miniblocks associated with λ_i, namely s_i.

Moreover, it is also possible to demonstrate that, under the assumption that the Jordan miniblocks associated with the same eigenvalue are arranged in decreasing dimension order, the **minimal (annihilating) polynomial** of the matrix F can be expressed as:

$$\Psi_F(z) = (z-\lambda_1)^{n_{1,1}}(z-\lambda_2)^{n_{2,1}} \dots (z-\lambda_r)^{n_{r,1}} = \prod_{i=1}^{r}(z-\lambda_i)^{n_{i,1}}.$$

As it will be remarked later on in this section, the minimal polynomial plays an important role in the modal analysis of a system. Finally, it is worth noticing that

$$n = \sum_{i=1}^{r} \mathrm{dim} F_i = \sum_{i=1}^{r}\sum_{\ell=1}^{s_i} \mathrm{dim} J_{i,\ell} = \sum_{i=1}^{r}\sum_{\ell=1}^{s_i} n_{i,\ell}.$$

The following proposition formalizes the importance of the Jordan form.

Proposition 1.4.5 *The matrix $F \in \mathbb{C}^{n\times n}$ is similar to a matrix in Jordan form, namely there exists a non singular matrix $T \in \mathbb{C}^{n\times n}$ such that $T^{-1}FT$ is in*

Jordan form. Without loss of generality, it will be assumed that the miniblocks of the matrix in Jordan form are arranged according to the aforementioned criterion.

Let us indicate with J the Jordan form of F. The matrices F and J represent, with respect to different basis, the same linear transformation, thus they have the same eigenvalues with the same (geometric and algebraic) multiplicities and the same minimal polynomial. Let T represent a similarity matrix such that

$$J = T^{-1}FT.$$

Then it holds

$$F^k = TJ^kT^{-1}, \qquad k \in \mathbb{Z}_+,$$

thus, once the expression of the generic k-th power of J is computed, the generic expression for F^k can be easily derived. In order to evaluate J^k, it is necessary to preliminary evaluate the k-th power of the generic Jordan miniblock, as shown by the following proposition.

Proposition 1.4.6 *Let λ be a generic complex number. Let us consider the Jordan miniblock of dimension ν associated with λ, namely the matrix*

$$J_\lambda = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{bmatrix} \in \mathbb{C}^{\nu\times\nu}.$$

Then, for $\lambda \neq 0$ and $k \in \mathbb{Z}_+$ it holds

$$(J_\lambda)^k = \begin{bmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \cdots & \cdots & \binom{k}{\nu-1}\lambda^{k-\nu+1} \\ & \lambda^k & \binom{k}{1}\lambda^{k-1} & & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & \lambda^k & \binom{k}{1}\lambda^{k-1} \\ & & & & \lambda^k \end{bmatrix}. \tag{1.13}$$

Moreover, for $\lambda = 0$ and $k \in \mathbb{Z}_+$, it holds [3]

$$
(J_0)^k = \begin{bmatrix}
\delta(k) & \delta(k-1) & \dots & \dots & \delta(k-\nu+1) \\
 & \delta(k) & \delta(k-1) & & \vdots \\
 & & \ddots & \ddots & \vdots \\
 & & & \delta(k) & \delta(k-1) \\
 & & & & \delta(k)
\end{bmatrix}. \tag{1.14}
$$

PROOF The matrix J_0 has a finite number of nonzero powers[4] since

$$
(J_0)^2 = \begin{bmatrix}
0 & 0 & 1 & \dots & 0 \\
 & 0 & 0 & 1 & \vdots \\
 & & \ddots & \ddots & 1 \\
 & & & 0 & 0 \\
 & & & & 0
\end{bmatrix}
\qquad
(J_0)^3 = \begin{bmatrix}
0 & 0 & 0 & 1 & \dots & & \dots & 0 \\
 & 0 & 0 & 0 & 1 & & & \vdots \\
 & & \ddots & \ddots & & \ddots & & \\
 & & & \ddots & \ddots & & \ddots & \vdots \\
 & & & & \ddots & \ddots & & 1 \\
 & & & & & 0 & 0 & 0 \\
 & & & & & & 0 & 0 \\
 & & & & & & & 0
\end{bmatrix}
$$

[3] The case $\lambda = 0$ can be interpreted as a special instance of the previous one. In fact, it is well known that $\binom{k}{\ell} = 0$ for $\ell > k$ and that all the k-th powers of 0 are null except for the 0-th one, that is equal to 1. Therefore for $\lambda = 0$, $\binom{k}{\ell}\lambda^{k-\ell} = \delta(k-\ell)$. However, for the sake of clarity, we will reserve a specific discussion for the case $\lambda = 0$.

[4] A matrix $M \in \mathbb{C}^{\nu\times\nu}$ such that there exists a positive integer number k such that $M^k = 0$ is said to be a **nilpotent matrix**. Moreover, for such a matrix, the smallest k such that $M^k = 0$, namely $\min\{k \in \mathbb{N} : M^k = 0\}$, is called **nilpotency index**. In this specific case, J_0 is a nilpotent matrix whose nilpotency index is ν.

A general characterization for nilpotent matrices is given in the following: $M \in \mathbb{C}^{\nu\times\nu}$ satisfies $M^k = 0$ if and only if $p(z) = z^k$ is an annihilating polynomial for the M matrix. Keeping in mind the relation between any annihilating polynomial and the minimum annihilating polynomial and the Cayley-Hamilton theorem, it immediately follows that for a matrix $M \in \mathbb{C}^{\nu\times\nu}$ the following statements are equivalent:

i) M is nilpotent;

ii) the minimal polynomial of M, $\psi_M(z)$, is of the type z^m, $\exists\ m \in \mathbb{N}$;

iii) the characteristic polynomial of M, $\Delta_M(z)$, is z^ν;

iv) all the eigenvalues of M are zero , i.e. $\sigma(M) = \{0\}$.

It is also worth noticing that the nilpotency index of M is equal to the degree m of the minimal polynomial $\psi_M(z)$: in fact, by definition of minimal polynomial, $\psi_M(M) = M^m = 0$, thus the nilpotency index is less than or equal to m. On the other hand, if the nilpotency index were an integer $k < m$, $p(z) = z^k$ would be an annihilating polynomial of degree lower than the one of the minimal polynomial, thus leading to a contradiction. Therefore m corresponds to the nilpotency index.

$$(J_0)^{\nu-1} = \begin{bmatrix} 0 & 0 & 0 & \dots & 1 \\ & 0 & 0 & 0 & \vdots \\ & & \ddots & \ddots & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{bmatrix} \qquad (J_0)^{\nu} = 0.$$

Consequently, keeping in mind that $(J_0)^0 = I_\nu$ and $(J_0)^1 = J_0$, it is possible to represent the generic k-th power of the matrix J_0 by means of discrete impulses, thus in the form (1.14).

Since, for $\lambda \neq 0$, it holds that

$$J_\lambda = \lambda I_\nu + J_0,$$

where λI_ν is a scalar matrix [5] and the matrices λI_ν and J_0 commute[6], we can exploit the Newton binomial formula in order to write the k-th power of the matrix $(J_\lambda)^k$, as

$$(J_\lambda)^k = (\lambda I_\nu + J_0)^k = \sum_{i=0}^{k} \binom{k}{i} (\lambda I_\nu)^{k-i} (J_0)^i = \sum_{i=0}^{k} \binom{k}{i} \lambda^{k-i} (J_0)^i.$$

Finally, it is worth noticing that, for $i \geq \nu$, it holds that $(J_0)^i = 0$. Thus, for any value of k it holds

$$(J_\lambda)^k = \sum_{i=0}^{\nu-1} \binom{k}{i} \lambda^{k-i} (J_0)^i.$$

By exploiting the expression for $(J_0)^i$ appearing in the first part of the proof, the identity (1.13) is obtained. ■

We are now in the position to determine the generic expression of the k-th power of a matrix F.

Proposition 1.4.7 *Assume that $F \in \mathbb{C}^{n\times n}$ has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_r$, and let $T \in \mathbb{C}^{n\times n}$ be a non singular matrix such that $T^{-1}FT$ is in Jordan form:*

$$J = T^{-1}FT = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_r \end{bmatrix}, \qquad \text{with} \quad J_i = \begin{bmatrix} J_{i,1} & & & \\ & J_{i,2} & & \\ & & \ddots & \\ & & & J_{i,s_i} \end{bmatrix},$$

with $J_{i,\ell}$ the ℓ-th Jordan miniblock associated with the eigenvalue λ_i, of dimension $n_{i,\ell}$. Then

$$F^k = TJ^kT^{-1} = T \begin{bmatrix} (J_1)^k & & & \\ & (J_2)^k & & \\ & & \ddots & \\ & & & (J_r)^k \end{bmatrix} T^{-1},$$

[5] A **scalar matrix** is a diagonal matrix whose diagonal elements are all identical. It is so called since it acts, with respect to the product with other matrices, as a scalar, since for any square matrix M it holds that $(\lambda I) \cdot M = M \cdot (\lambda I) = \lambda \cdot M$.

[6] Two square matrices M_1 an M_2 of the same dimensions, **commute** if $M_1 \cdot M_2 = M_2 \cdot M_1$.

where every matrix $(J_i)^k$ is a block diagonal matrix, whose ℓ-th diagonal block is a matrix (of dimension $n_{i,\ell}$) of the type

$$(J_{i,\ell})^k = \begin{bmatrix} \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \dots & \dots & \binom{k}{n_{i,\ell}-1}\lambda_i^{k-n_{i,\ell}+1} \\ & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} \\ & & & & \lambda_i^k \end{bmatrix} \tag{1.15}$$

if $\lambda_i \neq 0$, and of the type

$$(J_{i,\ell})^k = \begin{bmatrix} \delta(k) & \delta(k-1) & \dots & \dots & \delta(k-n_{i,\ell}+1) \\ & \delta(k) & \delta(k-1) & & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & \delta(k) & \delta(k-1) \\ & & & & \delta(k) \end{bmatrix} \tag{1.16}$$

if $\lambda_i = 0$.

Each of the elementary sequences $\binom{k}{\ell}\lambda_i^{k-\ell}$, associated with the eigenvalues $\lambda_i \in \mathbb{C} \setminus \{0\}$, and each of the impulses $\delta(k-\ell)$, associated with the null eigenvalue, is called **elementary mode** of the matrix F or, equivalently, of the system. Since the unforced evolution of a state space model can be written as $\mathbf{x}(k) = F^k\mathbf{x}_0$, for the state, and as $\mathbf{y}(k) = HF^k\mathbf{x}_0$, for the output, from the previous proposition it can be deduced that each of the entries $x_j(k)$ and $y_j(k)$ of the unforced state and output evolutions, respectively, is a linear combination of the elementary modes of the system. The combinatorial coefficients depend on the matrix T (and on its inverse) and on the initial conditions vector $\mathbf{x}_0$. However, there are some elementary modes that do not appear in the expression of the output unforced evolution, due to the presence of the H matrix.

Generally, in fact, just a subset of the elementary modes is “observable”, namely detectable from the analysis of the output unforced evolution, interpreted as “system observation”. We will come back to this concept later on.

If F is a matrix with real coefficients then its complex eigenvalues (if any) appear in conjugate pairs. Moreover, it is possible to demonstrate that if in the Jordan form of F there exists a Jordan miniblock of dimension ν associated with the complex eigenvalue $\lambda = \rho e^{j\theta}$ (with $\rho, \theta \in \mathbb{R}$), then there exist another Jordan miniblock of dimension ν associated with the complex conjugate eigenvalue $\overline{\lambda} = \rho e^{-j\theta}$. Consequently, if F is a real matrix, its complex modes appear in complex conjugate pairs and they are also combined with complex conjugate coefficients, so that the final expression is real. Therefore, it is always possible to replace the pair of complex conjugate modes $(\lambda^k, \overline{\lambda}^k)$ with the pair of real modes $(\rho^k \cos(\theta k), \rho^k \sin(\theta k))$. The same holds true for the complex modes of the type $\binom{k}{\ell}\lambda^{k-\ell}$ and $\binom{k}{\ell}\overline{\lambda}^{k-\ell}$, which can be replaced by the equivalent pair $\Big(\binom{k}{\ell}\rho^{k-\ell}\cos\Big(\theta(k-\ell)\Big), \binom{k}{\ell}\rho^{k-\ell}\sin\Big(\theta(k-\ell)\Big)\Big)$. In the following, we propose an equivalent and less compact writing for the expression of the elementary modes of a discrete time model. For $\lambda \neq 0$ and $k \geq \ell$, it holds,

$$\binom{k}{\ell}\lambda^{k-\ell} = \frac{k(k-1)\dots(k-\ell+1)}{\ell!\ \lambda^\ell}\lambda^k \delta_{-1}(k-\ell) = p_\ell(k)\ \lambda^k \delta_{-1}(k-\ell),$$

where

$$p_\ell(k) = \frac{k(k-1)\dots(k-\ell+1)}{\ell!\ \lambda^\ell}$$

is a polynomial in the time variable k of degree ℓ. However, the family of discrete exponential sequences

$$\lambda^k,\ k\ \lambda^k,\ \dots,\ \frac{k^{\mu-1}}{(\mu-1)!}\lambda^k,$$

and the family of elementary modes

$$\lambda^k, \binom{k}{1}\ \lambda^{k-1}, \dots, \binom{k}{\mu-1}\lambda^{k-\mu+1}$$

are equivalent for $k \in \mathbb{Z}_+$, since the linear combinations of the two families lead to the same sequences. It is worth noticing that the binomial coefficient appearing in the elementary modes expression brings information about the time instant at which the modes become "active". So, if we resort to the family of discrete exponential sequences we should also make use of appropriate discrete unitary steps.

To conclude we want to understand what are the distinct elementary modes involved in the expression of the unforced state evolution (and of the output unforced evolution as well even, if some of them may be weighted with a zero coefficient), as $\mathbf{x}_0$ in $\mathbb{R}^n$ varies.

As it can be deduced from (1.15)-(1.16), every Jordan miniblock $J_{i,\ell}$ of dimension $n_{i,\ell}$ associated with the eigenvalue λ_i corresponds to $n_{i,\ell} = \dim J_{i,\ell}$ distinct and linearly independent elementary modes. Obviously, when taking into account two Jordan miniblocks associated with the same eigenvalue λ_i, the number of distinct elementary modes identified by the two miniblocks coincides with the dimension of the largest of the miniblocks. To sum up, the maximum number of distinct elementary modes associated with the eigenvalue λ_i corresponds to the dimension of the largest Jordan miniblock associated with λ_i and hence, assuming that the Jordan miniblocks arrangement is ordered, with $n_{i,1}$.

The overall number of distinct elementary modes associated with the matrix F is $\sum_{i=1}^{r} n_{i,1}$. It is worth noticing that the information about the number and the expression of the system modes is given by the minimal polynomial of the matrix F since, as mentioned before, it can be written as

$$\Psi_F(z) = \prod_{i=1}^{r} (z - \lambda_i)^{n_{i,1}}.$$

We want now to analyze the (convergent, limited or divergent) behaviour of the elementary modes for $k \in \mathbb{Z}_+$.

Proposition 1.4.8 *The elementary mode* $m(k) = \binom{k}{\ell}\lambda^{k-\ell}, k, \ell \in \mathbb{Z}_+$, *and* $\lambda \in \mathbb{C} \setminus \{0\}$, *is*

- *convergent (to zero, for* $k \to +\infty$*), namely*

$$\lim_{k \to +\infty} m(k) = 0,$$

  if and only if $|\lambda| < 1$*;*

- *limited, namely* $\exists\ M \in \mathbb{R}_+$ *such that*

$$|m(k)| < M, \qquad \forall\ k \in \mathbb{Z}_+,$$

  if and only if $|\lambda| \leq 1$ *and, when* λ *has unitary modulus,* $\ell = 0$*;*

- *divergent (for* $k \to +\infty$*) otherwise.*

The elementary modes associated with the null eigenvalue (if any) $\delta(k - \ell)$*, with* $k, \ell \in \mathbb{Z}_+$*, are always convergent to zero (in a finite number of steps).*

1.5 System Analysis in the Z-Transform Domain

Let us consider a discrete time LTI state space model, described by the equations

$$\mathbf{x}(k+1) = F\mathbf{x}(k) + G\mathbf{u}(k), \qquad (1.17)$$

$$\mathbf{y}(k) = H\mathbf{x}(k) + D\mathbf{u}(k), \qquad k \in \mathbb{Z}_+. \qquad (1.18)$$

We assume that the system evolves starting from the time instant $k = 0$ from an arbitrary initial condition $\mathbf{x}(0) = \mathbf{x}_0$, and the input $\mathbf{u}(k)$ is known for $k \in \mathbb{Z}_+$. Moreover we suppose that the input sequence applied to the system admits $\mathcal{Z}$-transform (see Appendix), by this meaning that each component of the vector sequence admits $\mathcal{Z}$-transform. The $\mathcal{Z}$-transform of the input vector sequence $\mathbf{u}(k)$ is a vector whose i-th component is the $\mathcal{Z}$-transform of the i-th component of the vector $\mathbf{u}(k)$, namely $u_i(k)$. Under this assumption, it is possible to demonstrate that also the state and output sequences admit $\mathcal{Z}$-transforms and therefore we can assume

$$U(z) \doteq \mathcal{Z}[\mathbf{u}(k)], \qquad X(z) \doteq \mathcal{Z}[\mathbf{x}(k)], \qquad Y(z) \doteq \mathcal{Z}[\mathbf{y}(k)].$$

By applying the $\mathcal{Z}$-transform to both sides of the two equations (1.17) and (1.18) and by using the left shift property of the $\mathcal{Z}$-transform (Property 4)) we get

$$\begin{aligned} zX(z) - z\mathbf{x}_0 &= FX(z) + GU(z), \\ Y(z) &= HX(z) + DU(z), \end{aligned}$$

from which it immediately follows

$$\begin{aligned} X(z) &= (zI_n - F)^{-1} z\mathbf{x}_0 + (zI_n - F)^{-1} GU(z), && (1.19) \\ Y(z) &= HX(z) + DU(z) \\ &= H(zI_n - F)^{-1} z\mathbf{x}_0 + \left(H(zI_n - F)^{-1} G + D\right) U(z). && (1.20) \end{aligned}$$

We notice that both the $\mathcal{Z}$-transform of the state and of the output consist of two terms: one depending on the initial condition $\mathbf{x}_0$ (and not on the input) and one depending on the input $\mathcal{Z}$-transform (and not on the initial condition $\mathbf{x}_0$). Then, upon indicating by $\mathbf{x}_\ell$ and $\mathbf{y}_\ell$ the state and input unforced evolution components, respectively, and by $\mathbf{x}_f$ and $\mathbf{y}_f$ the state and input forced evolution components, respectively, we get

$$\begin{aligned} X_\ell(z) &\doteq \mathcal{Z}[\mathbf{x}_\ell(k)] = (zI_n - F)^{-1} z\mathbf{x}_0, \\ Y_\ell(z) &\doteq \mathcal{Z}[\mathbf{y}_\ell(k)] = H(zI_n - F)^{-1} z\mathbf{x}_0, \\ X_f(z) &\doteq \mathcal{Z}[\mathbf{x}_f(k)] = (zI_n - F)^{-1} GU(z), \\ Y_f(z) &\doteq \mathcal{Z}[\mathbf{y}_f(k)] = \left(H(zI_n - F)^{-1} G + D\right) U(z). \end{aligned}$$

The rational matrix

$$W(z) \doteq H(zI_n - F)^{-1} G + D \tag{1.21}$$

is called **transfer matrix** of the state space model. Notice that $W(z)$ is the $\mathcal{Z}$-transform of the impulse response of the system, namely

$$W(z) = \mathcal{Z}[W(k)].$$

Finally, it is worth noticing that

$$W(z) = \frac{1}{\det(zI_n - F)} \cdot H\mathrm{adj}(zI_n - F)G + D = \frac{1}{\Delta_F(z)} \cdot H\mathrm{adj}(zI_n - F)G + D,$$

where $\Delta_F(z)$ is a monic polynomial of degree n, while $\mathrm{adj}(zI_n - F)$ is the adjoint matrix of $zI_n - F$ and, by its own definition, is a polynomial matrix of degree $n-1$

(recall that the (i, j)-th element of $\text{adj}(zI_n - F)$ is the algebraic complement of the (j, i)-th element of $zI_n - F$). Therefore $W(z)$ is a proper rational matrix, that can be decomposed in the sum of a strictly proper rational component, $H(zI_n - F)^{-1}G$, and a constant term D. Its dimension is $p \times m$, since it relates the $\mathcal{Z}$-transform of the input to the one of the output forced component. The region of convergence of $W(z)$ (resulting from the intersection of the regions of convergence of its components) is $|z| > |p|$, where p is the pole of maximum modulus among the poles of $W(z)$ (equivalently, among the poles of its components).

One may wonder what relation there exists between the poles of $W(z)$ and the eigenvalues of F. In order to address this question we write $W(z)$ in the form

$$W(z) = \frac{1}{\Delta_F(z)} \cdot H\text{adj}(zI_n - F)G + D$$

from which it is immediate to notice that the poles of $W(z)$ are a subset of the zeros of $\Delta_F(z)$, namely the eigenvalues of F. In fact, some simplifications between the polynomial $\Delta_F(z)$, appearing as denominator, and all the polynomials appearing as numerators (namely the entries of $H\text{adj}(zI_n - F)G$) may occur. In this case, one or more eigenvalues may be canceled as zeros of the denominator of $W(z)$. Therefore, generally, it holds that

$$\{\text{poles of } W(z)\} \subseteq \{\text{eigenvalues of } F\}.$$

1.6 Discrete Time State Space Model Stability

In this section we introduce some definitions of the stability for state space models. The first two definitions (one more restrictive than the other) refer to the unforced state evolution, the third one, instead, refers to the forced output evolution. Due to the different areas of interest, we will refer to the first two notions as **internal stability** forms and to the last one as **external or BIBO stability** (where the acronym stands for Bounded Input Bounded Output) of the state space models.

Definition 1.6.1 *The system (1.17)÷(1.18) is said to be*

- **asymptotically stable** *if, for any choice of the initial condition* $\mathbf{x}(0)$*, the system unforced state evolution* $\mathbf{x}_\ell(k)$ *asymptotically converges to zero , i.e.*

$$\lim_{k \to +\infty} \mathbf{x}_\ell(k) = 0;$$

- **simply stable** *if, for any choice of the initial condition* $\mathbf{x}(0)$*, the system unforced state evolution* $\mathbf{x}_\ell(k)$ *is a limited function, by this meaning that there exists* $M \in \mathbb{R}_+$ *such that*

$$\|\mathbf{x}_\ell(k)\| \leq M \qquad \text{for every } k \in \mathbb{Z}_+;$$

- **unstable** *otherwise.*

Moreover, the system (1.17)÷(1.18) is said to be **externally or BIBO (bounded input/bounded output) stable** *if, starting from zero initial conditions, it responds to every input (zero for $k < 0$ and) bounded for $k \geq 0$ with a bounded (forced) output. In other words, by setting $\mathbf{x}_0 = 0$, for every input $\mathbf{u}(k), k \in \mathbb{Z}_+$, for which there exists $M_u > 0$ such that $\|\mathbf{u}(k)\| < M_u$ for every $k \in \mathbb{Z}_+$, it is possible to determine M_v such that $\|\mathbf{y}(k)\| = \|\mathbf{y}_f(k)\| < M_v$ for every $k \in \mathbb{Z}_+$.*

By making use of the detailed analysis made for the derivation of the unforced state evolution, we are now in the position to provide a comprehensive characterization of the two aforementioned internal stability properties for state space models. In fact, since

$$\mathbf{x}_\ell(k) = TJ^kT^{-1}\mathbf{x}_0,$$

where J is the Jordan form of F and T is a basis change matrix that allows to derive J from F (and vice versa), the only case in which $\mathbf{x}_\ell(k)$ always converge to zero, regardless of $\mathbf{x}_0$, occurs when all the elementary sequences appearing in J^k, namely all the elementary modes of the system, converge to zero.

Analogously, the only case when $\mathbf{x}_\ell(k)$ is always bounded, regardless of $\mathbf{x}_0$, occurs when all the elementary sequences appearing in J^k, namely all the elementary modes of the system, are bounded. It is worth noticing that when there is an eigenvalue of unitary modulus with more than one elementary modes associated with it, then there exists a "first" bounded elementary mode corresponding to it, while the "remaining" ones are divergent modes. Therefore, the only case in which an eigenvalue with unitary modulus is associated only with bounded modes, is when in the Jordan form of F such eigenvalue only appears in miniblocks of dimension one. This occurs if and only if the largest among the miniblocks associated with such a eigenvalue in the Jordan form of F has unitary dimension or, equivalently, the multiplicity of the eigenvalue in the minimal polynomial $\Psi_F(z)$ is equal to one.

Therefore, by resorting to the analysis of the behaviour of the elementary modes, provided in Proposition 1.4.8, we get the following characterization of the asymptotic and simple stability properties for a discrete time state space model.

Proposition 1.6.2 *The system (1.17)÷(1.18) is*

- *asymptotically stable if and only if all the zeros of the characteristic polynomial of the system $\Delta_F(z)$ (equivalently, of the minimal polynomial of the system $\Psi_F(z)$) have moduli smaller than 1, namely all the eigenvalues of F have moduli smaller than 1;*

- *simply stable if and only if all the zeros of the minimal polynomial of the system $\Psi_F(z)$ have moduli smaller than or equal to 1 and every zero with unitary modulus has unitary multiplicity, equivalently, all the eigenvalues of F have moduli smaller than or equal to 1 and the eigenvalues with unitary modulus are associated, in the Jordan form of F, to miniblocks of unitary dimension.*

It is worth noticing that, for the study of the asymptotic stability of the system, one can equivalently look at the zeros of the characteristic polynomial or of the minimal polynomial. On the other hand, when one is interested in the simple stability of the system, the only analysis of the characteristic polynomial, in general, is not sufficient. It is sufficient only when either there exist unstable zeros belonging to $\Delta_F(z)$ (in this situation the system is not simple stable since such unstable zeros appear in the minimal polynomial as well) or all the zeros of $\Delta_F(z)$ have moduli less than or equal to one and the zeros with unitary moduli have unitary (algebraic) multiplicity (in this case the same holds for the minimal polynomial, thus ensuring the simple stability of the system). The case in which the zeros of $\Delta_F(z)$ have moduli smaller than or equal to one, but there exists at least one zero with unitary modulus and algebraic multiplicity greater than one, requires a more in-depth analysis since the minimal polynomial (or at least the multiplicity of the "critical" eigenvalues) needs to be inspected. We will further stress this concept through some examples.

As far as BIBO stability is concerned, the following characterization holds. The proof is omitted.

Proposition 1.6.3 *Given a discrete time system described as in (1.17)÷(1.18), the following facts are equivalent:*

i) the state space model is BIBO stable;

ii) the norm of the impulse response of the system $W(k), k \in \mathbb{Z}_+$, is summable, namely

$$\sum_{k=0}^{+\infty} \|W(k)\| < \infty; \tag{1.22}$$

iii) all entries $w_{ij}(k)$ of the impulse response of the system $W(k), k \in \mathbb{Z}_+$, are summable in norm, namely

$$\sum_{k=0}^{+\infty} |w_{ij}(k)| < \infty, \qquad \forall\ i, j; \tag{1.23}$$

iv) all the elementary modes appearing (i.e., weighted by a nonzero coefficient) in the impulse response $W(k), k \in \mathbb{Z}_+$, are convergent;

v) all poles of the system transfer matrix $W(z)$ have moduli smaller than one (equivalently, the unitary circle belongs to the region of convergence of $W(z)$).

What is the relation between internal and external stability? As previously said, asymptotic stability of a state space model is related to the position of the zeros of the characteristic polynomial $\Delta_F(z)$ in the complex plane. On the other hand, BIBO stability is fully determined by the poles of the transfer function $W(z)$. However, the poles of $W(z)$ generally represent a subset of the zeros of $\Delta_F(z)$. Indeed, the poles of $W(z)$ are the zeros of the polynomial appearing as denominator of $W(z)$ in an irreducible representation, and it is not guaranteed that the representation of the transfer function given by

$$W(z) = \frac{1}{\Delta_F(z)} \cdot H\mathrm{adj}(zI_n - F)G + D$$

is an irreducible representation. It immediately follows that asymptotic stability always implies BIBO stability while the contrary does not hold, since cancellations in the aforementioned representation of $W(z)$ may occur. In fact, if each "unstable" zero λ of $\Delta_F(z)$ (namely each zero with $|\lambda| \geq 1$) is a zero of all the entries of $H\mathrm{adj}(zI_n - F)G$ and the multiplicity of λ as zero of $\Delta_F(z)$ is not greater than its multiplicity as a zero of all the entries of $H\mathrm{adj}(zI_n - F)G$, then the system is BIBO stable although it is not asymptotically stable.

Clearly, a similar discussion could be carried out in the time domain. We will underline this concept later in the course, by showing the relation between zero-pole cancellations and the non-minimality of the state space model represented by its transfer matrix, due to the presence of "not reachable" and/or "not observable" eigenvalues.

Example 1.6.4 Let us consider the following state space model:

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + G\mathbf{u}(k) = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathbf{u}(k), \\ \mathbf{y}(k) &= H\mathbf{x}(k) = [\,1 \quad 1\,]\,\mathbf{x}(k), \qquad k \in \mathbb{Z}_+. \end{aligned}$$

The characteristic polynomial of the matrix F is $\Delta_F(z) = (z-2)(z-1/2)$. Therefore, F has an eigenvalue of modulus greater than 1. This implies that the system is neither asymptotically stable nor simply stable. In order to test BIBO stability it suffices to evaluate the transfer function of the system and its poles. One finds

$$W(z) = \frac{1}{z - 1/2},$$

from which it follows that the system is BIBO stable. ♠

Example 1.6.5 Let us consider the following state space model:

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + G\mathbf{u}(k) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mathbf{u}(k), \\ \mathbf{y}(k) &= H\mathbf{x}(k) = [\,1 \quad 0 \quad 1\,]\,\mathbf{x}(k), \qquad k \in \mathbb{Z}_+. \end{aligned}$$

The characteristic polynomial of the matrix F is $\Delta_F(z) = (z-1)^2 z$. Thus, F has a simple null eigenvalue and an eigenvalue of unitary modulus with algebraic multiplicity equal to 2. Obviously, the system is not asymptotically stable. In order to test its simple stability, we need to evaluate the dimensions of the Jordan miniblock/miniblocks related to the eigenvalue 1. Since the algebraic multiplicity of the eigenvalue 1 is 2, the sum of the dimensions of the miniblocks related to the eigenvalue 1 is 2. Therefore only two cases may occur: a) there are two Jordan miniblocks of dimension one associated with the unitary eigenvalue (thus the system is simply stable), b) there is one Jordan miniblock of dimension 2 associated with the unitary eigenvalue (thus the system is unstable).

How can we understand which one of the two cases occurs? Since the number of miniblocks associated with a given eigenvalue coincides, as already said, with the geometric multiplicity of the eigenvalue, namely with the dimension of the eigenspace $\ker(\lambda I - F)$, case a) corresponds to the case when $\dim(\ker(1 \cdot I_3 - F)) = 2$, while case b) to the case when $\dim(\ker(1 \cdot I_3 - F)) = 1$. We notice that

$$\ker(1 \cdot I_3 - F) = \ker\left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right) = \langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rangle,$$

where $\langle \mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_l \rangle$ represents the vector subspace generated by the vectors $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_l$. Since $\dim(\ker(1 \cdot I_3 - F)) = 2$, the system is simply stable. Let us focus now on BIBO stability. The computation of the transfer function leads to $W(z) = \frac{1}{z-1}$, from which it follows that the system is not BIBO stable. ♠

1.7 The Tank Example (contd.)

Let us go back to the tank example that we exploited to introduce the discrete time state space model. Let us focus now on the dynamics of this model. The characteristic polynomial of the matrix F is

$$\begin{aligned} \Delta_F(z) &= \det \begin{bmatrix} z - (1-a) & -a \\ b & z-1 \end{bmatrix} = (z - 1 + a)(z - 1) + ab \\ &= z^2 + (a-2)z + (ab + 1 - a). \end{aligned}$$

The eigenvalues of F are the zeros of its characteristic polynomial, namely

$$\lambda_{1,2} = \frac{2 - a \pm \sqrt{(a-2)^2 - 4(ab + 1 - a)}}{2} = \frac{2 - a \pm \sqrt{a(a-4b)}}{2}.$$

Let us suppose that $a - 4b \neq 0$ (a is different from zero in this example). In this case we have two distinct roots, λ_1 and λ_2, which are real (if $a - 4b > 0$) or complex conjugate (if $a - 4b < 0$). Thus, the Jordan form of the matrix F is

$$J = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

Moreover, since the matrix F is diagonalizable, the matrix T relating F to J through the relation $F = TJT^{-1}$ is a matrix having as columns two eigenvectors $\mathbf{y}_1$ and $\mathbf{y}_2$, corresponding to λ_1 and λ_2, respectively. It is easy to verify that

$$T = \begin{bmatrix} \lambda_1 - 1 & \lambda_2 - 1 \\ -b & -b \end{bmatrix},$$

is a possible transformation matrix. Its inverse takes the form

$$T^{-1} = \frac{1}{b(\lambda_2 - \lambda_1)} \begin{bmatrix} -b & 1 - \lambda_2 \\ b & \lambda_1 - 1 \end{bmatrix}.$$

The system state updating equation can be thus rewritten as

$$\mathbf{x}(k+1) = (TJT^{-1})\mathbf{x}(k) + Gu(k) = T \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} T^{-1}\mathbf{x}(k) + \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^*$$

and the expression of the state at the generic time instant $k \in \mathbb{Z}_+$ is given by

$$\begin{aligned} \mathbf{x}(k) &= T \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} T^{-1}\mathbf{x}(0) + \sum_{i=0}^{k-1} T \begin{bmatrix} \lambda_1^{k-1-i} & 0 \\ 0 & \lambda_2^{k-1-i} \end{bmatrix} T^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^* \\ &= T \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} T^{-1}\mathbf{x}(0) + T \begin{bmatrix} \frac{\lambda_1^k - 1}{\lambda_1 - 1} & 0 \\ 0 & \frac{\lambda_2^k - 1}{\lambda_2 - 1} \end{bmatrix} T^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^*. \end{aligned}$$

It is worth investigating, now, if and under what conditions we are able to ensure that the temperature of the water in the tank asymptotically converges to the desired value T_S^*, regardless of the initial conditions $T_S(0)$ and $T_M(0)$. Let us observe, as a first step, that if $\lim_{k\to+\infty} T_S(k)$ exists and such a limit is equal to T_S^*, then $\lim_{k\to+\infty} T_M(k)$ exists and such a limit equals T_S^* as well, since

$$\lim_{k\to+\infty} T_M(k) = \lim_{k\to+\infty} \frac{T_S(k+1) - (1-a)T_S(k)}{a} = T_S^*.$$

Notice that this result was predictable, in fact this occurrence physically corresponds to the reaching of the thermal equilibrium between the water in the tank and the water provided by the mixer. In order to guarantee the existence of

$$\lim_{k\to+\infty} \mathbf{x}(k) = \begin{bmatrix} T_S^* \\ T_S^* \end{bmatrix},$$

regardless of the initial conditions, it is necessary that the effect of the initial condition asymptotically converge to the zero vector. If this were not the case, in fact, distinct initial conditions would lead to distinct limits, since the unforced evolution components in the expression of $\mathbf{x}(k)$ would differ, while the forced evolution components would coincide.

This amounts to saying that it must be $|\lambda_i| < 1$ for $i = 1, 2$. Simple calculations allow to demonstrate that the system is asymptotically stable if and only if $b < 1$.

If the system is asymptotically stable, then, from the previous expression of $\mathbf{x}(k)$, one gets

$$\begin{aligned}
\lim_{k\to+\infty} \mathbf{x}(k) &= T \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} T^{-1}\mathbf{x}(0) + T \begin{bmatrix} \frac{\lambda_1^k-1}{\lambda_1-1} & 0 \\ 0 & \frac{\lambda_2^k-1}{\lambda_2-1} \end{bmatrix} T^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^* \\
&= T \begin{bmatrix} \frac{-1}{\lambda_1-1} & 0 \\ 0 & \frac{-1}{\lambda_2-1} \end{bmatrix} T^{-1} \begin{bmatrix} 0 \\ b \end{bmatrix} T_S^*.
\end{aligned}$$

In order to verify that the limit thus obtained coincides with $\begin{bmatrix} T_S^* \\ T_S^* \end{bmatrix}$, it suffices to replace the matrices T and T^{-1} and the roots, λ_1 and λ_2 with the previously determined expressions.

1.8 Exercises and Examples

Example 1.8.1 Let us consider the discrete time system

$$\mathbf{x}(k+1) = F\mathbf{x}(k), \qquad k \in \mathbb{Z}_+,$$

whose system matrix F is in Jordan form, specifically

$$F = \left[\begin{array}{cc|c|ccc|c}
\lambda_1 & 1 & & & & & \\
0 & \lambda_1 & & & & & \\
\hline
 & & \lambda_1 & & & & \\
\hline
 & & & \lambda_2 & 1 & 0 & \\
 & & & 0 & \lambda_2 & 1 & \\
 & & & 0 & 0 & \lambda_2 & \\
\hline
 & & & & & & \lambda_3
\end{array}\right],$$

where λ_1, λ_2 and λ_3 are arbitrary (not necessarily distinct) real numbers. Determine, as λ_1, λ_2 and λ_3 vary in $\mathbb{R}$

a) the characteristic polynomial of F;

b) the minimal polynomial and the elementary modes of the system;

c) the dimensions of the distinct eigenspaces $\ker(\lambda_i I_7 - F)$, $i = 1, 2, 3$.

SOLUTION The characteristic polynomial of the matrix, can always be expressed, regardless of the values of $\lambda_1, \lambda_2, \lambda_3$, as

$$\Delta_F(z) = (z - \lambda_1)^3 (z - \lambda_2)^3 (z - \lambda_3),$$

where the integers $3, 3$ and 1 represent the algebraic multiplicities of the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ only if they are distinct. As far as the other two questions are concerned, we have to distinguish different cases, depending on whether $\lambda_1, \lambda_2, \lambda_3$ are distinct eigenvalues, only two of them coincide or all of them coincide. Consequently, we can distinguish the following cases:

1. $\lambda_1 = \lambda_2 = \lambda_3$;

2. $\lambda_1 = \lambda_2 \neq \lambda_3$;

3. $\lambda_1 \neq \lambda_2 = \lambda_3$;

4. $\lambda_1 = \lambda_3 \neq \lambda_2$;

5. $\lambda_1 \neq \lambda_2 \neq \lambda_3$.

1. The first case corresponds to a Jordan form with 4 miniblocks associated with the same eigenvalue, say λ. The largest miniblock has dimension 3. Therefore $\Psi_F(z) = (z-\lambda)^3$, and the modes are

$$\lambda^k, \binom{k}{1}\lambda^{k-1}, \binom{k}{2}\lambda^{k-2}.$$

The dimension of the eigenspace $\ker(\lambda I_7 - F)$ coincides with the number of miniblocks associated with λ, and hence it is equal to 4.

2. The second case corresponds to a Jordan form with 3 miniblocks associated with the same eigenvalue $\lambda_1 = \lambda_2$, let us call it λ. The largest miniblock has dimension 3, and there exists only one miniblock of dimension 1 associated with λ_3. Thus $\Psi_F(z) = (z-\lambda)^3(z-\lambda_3)$, the modes are

$$\lambda^k, \binom{k}{1}\lambda^{k-1}, \binom{k}{2}\lambda^{k-2}, \lambda_3^k.$$

The dimension of the eigenspace $\ker(\lambda I_7 - F)$ coincides with the number of miniblocks associated with λ and hence it is 3, the dimension of the eigenspace $\ker(\lambda_3 I_7 - F)$ coincides with the number of miniblocks associated with λ_3 and hence it is 1.

3. The third case corresponds to a Jordan form with two miniblocks associated with the same eigenvalue $\lambda_2 = \lambda_3$, let us call it λ. The largest miniblock has dimension 3. There are two miniblocks associated with λ_1, the largest of the two being of dimension 2. Thus $\Psi_F(z) = (z-\lambda)^3(z-\lambda_1)^2$, the modes are

$$\lambda^k, \binom{k}{1}\lambda^{k-1}, \binom{k}{2}\lambda^{k-2}, \lambda_1^k, \binom{k}{1}\lambda_1^{k-1}.$$

The dimension of the eigenspace $\ker(\lambda I_7 - F)$ coincides with the number of miniblocks associated with λ and thus it is 2, the dimension of the eigenspace $\ker(\lambda_1 I_7 - F)$ coincides with the number of miniblocks associated with λ_1 thus it is equal to 2.

4. The fourth case corresponds to a Jordan form with 2 miniblocks associated with the same eigenvalue $\lambda_1 = \lambda_3$, let us call it λ. The largest of the two miniblocks has dimension 3. There exists a miniblock of dimension 3 associated with λ_2. Thus $\Psi_F(z) = (z-\lambda)^2(z-\lambda_2)^3$, the modes are

$$\lambda_2^k, \binom{k}{1}\lambda_2^{k-1}, \binom{k}{2}\lambda_2^{k-2}, \lambda^k, \binom{k}{1}\lambda^{k-1}.$$

The dimension of the eigenspace $\ker(\lambda I_7 - F)$ coincides with the number of miniblocks associated with λ and hence is 3, the dimension of the eigenspace $\ker(\lambda_2 I_7 - F)$ coincides with the number of miniblocks associated with λ_2 and hence is 1.

5. The last case corresponds to a Jordan form with 2 miniblocks associated with the eigenvalue λ_1, the largest of the two having dimension 2. There exist a miniblock

of dimension 3 associated with λ_2, and a miniblock of dimension 1 associated with λ_3. Thus $\Psi_F(z) = (z-\lambda_1)^2(z-\lambda_2)^3(z-\lambda_3)$, the elementary modes are

$$\lambda_1^k, \binom{k}{1}\lambda_1^{k-1}, \lambda_2^k, \binom{k}{1}\lambda_2^{k-1}, \binom{k}{2}\lambda_2^{k-2}, \lambda_3^k.$$

The dimension of the eigenspace $\ker(\lambda_1 I_7 - F)$ is 2, the dimension of the eigenspace $\ker(\lambda_2 I_7 - F)$ is 1 and, finally, the dimension of the eigenspace $\ker(\lambda_3 I_7 - F)$ is 1. ♠

Example 1.8.2 Let us consider the discrete time model described by the following equations:

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + Gu(k) &&= \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 2 \\ 2 \end{bmatrix} u(k), \\ y(k) &= H\mathbf{x}(k) &&= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}(k), \qquad\qquad k \in \mathbb{Z}_+. \end{aligned}$$

i) Determine, by operating in the time domain, the state evolution of the system corresponding to the following conditions:

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad u(k) = \begin{cases} 5, & 0 \le k \le 5, \\ 0, & k > 5. \end{cases}$$

ii) Compute the transfer function of the system.

iii) Study the simple, asymptotic and BIBO stability of the system.

Solution i) It is immediate to notice that

$$F^2 = \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} = -F, \qquad F^3 = F \cdot F^2 = F \cdot (-F) = -F^2 = F,$$

therefore the generic power of the matrix F is

$$F^k = (-1)^{k+1} F, \qquad k \ge 1.$$

For $k \le 5$ the expression of the state evolution is given by

$$\begin{aligned} \mathbf{x}(k) &= F^k \mathbf{x}(0) + \sum_{i=0}^{k-1} F^{k-1-i} G u(i) = (-1)^{k+1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 5 \cdot \sum_{i=0}^{k-1} (-1)^{k-i} \begin{bmatrix} -2 \\ -2 \end{bmatrix} \\ &= (-1)^{k+1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 5 \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} \cdot \sum_{i=1}^{k} (-1)^i \\ &= \begin{cases} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 5 \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 11 \\ 11 \end{bmatrix} & \text{for } k = 1, 3, 5, \\ -\begin{bmatrix} 1 \\ 1 \end{bmatrix} & \text{for } k = 2, 4. \end{cases} \end{aligned}$$

For $k > 5$ the system evolves as pure unforced evolution and it holds

$$x(k) = F^{k-5}\mathbf{x}(5) = (-1)^{k-4} F \begin{bmatrix} 11 \\ 11 \end{bmatrix} = (-1)^{k-3} \begin{bmatrix} 11 \\ 11 \end{bmatrix}.$$

ii) The transfer function of the system is

$$W(z) = H(zI_2 - F)^{-1} G = [1 \quad 0] \begin{bmatrix} z-1 & 2 \\ -1 & z+2 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \frac{2}{z+1}.$$

iii) The characteristic polynomial of the matrix F is

$$\Delta_F(z) = z(z+1),$$

therefore the system is simply stable but it is not asymptotically stable. On the other hand, the transfer function of the system has a pole in -1 and therefore the system is not BIBO stable. ♠

Example 1.8.3 Let us consider the discrete time system described by the following equation

$$\mathbf{x}(k+1) = F\mathbf{x}(k) + Gu(k), \qquad k \in \mathbb{Z}_+,$$

where

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \qquad \text{e} \qquad G = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

i) Determine the Jordan form of the matrix F, the elementary modes of the system and their (convergent, limited, unlimited) behaviour;

ii) Determine, by making use of the $\mathcal{Z}$-transform, the forced state evolution corresponding to the input signal

$$u(k) = \begin{cases} 1, & k = 0, \\ -3, & k = 1, \\ 0, & k > 1. \end{cases}$$

Solution i) The computation of the characteristic polynomial of the system leads to

$$\begin{aligned} \Delta_F(z) &= \det \begin{bmatrix} z & -1 & 0 \\ 0 & z-2 & -1 \\ 0 & -1 & z-2 \end{bmatrix} = z \cdot \det \begin{bmatrix} z-2 & -1 \\ -1 & z-2 \end{bmatrix} = z[(z-2)^2 - 1] \\ &= z(z-1)(z-3), \end{aligned}$$

where we exploited the fact that the matrix F (and thus $zI_3 - F$) is block triangular. Since all the eigenvalues of the matrix $\lambda_1 = 0, \lambda_2 = 1, \lambda_2 = 3$ are simple, the matrix F is diagonalizable and its Jordan form is

$$J = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The elementary modes of the system are: $\delta(k)$, associated with λ_1, the constant sequence that is equal to 1 at each time instant $k \in \mathbb{Z}$ and the sequence $3^k, k \in \mathbb{Z}$. The first mode is convergent (in a finite number of steps), the second one is limited (but not convergent), the third one is divergent.

ii) As a first step let us evaluate the $\mathcal{Z}$-transform of the input sequence. Since it is a sequence with compact support, by exploiting the definition of $\mathcal{Z}$-transform one gets

$$U(z) = 1 \cdot z^{-0} + (-3) \cdot z^{-1} = 1 - 3z^{-1} = \frac{z-3}{z}.$$

The $\mathcal{Z}$-transform of the forced state evolution can be expressed in the form

$$X_f(z) = (zI_3 - F)^{-1} G U(z).$$

Since

$$\begin{aligned}
(zI_3 - F)^{-1} G U(z) &= \frac{1}{\Delta_F(z)} \operatorname{adj}(zI_3 - F) G U(z) \\
&= \frac{1}{\Delta_F(z)} \operatorname{adj}\left(\begin{bmatrix} z & -1 & 0 \\ 0 & z-2 & -1 \\ 0 & -1 & z-2 \end{bmatrix} \right) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{z-3}{z} \\
&= \frac{1}{z(z-1)(z-3)} \begin{bmatrix} * & * & 1 \\ * & * & z \\ * & * & z(z-2) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \frac{z-3}{z} = \begin{bmatrix} \frac{1}{z^2(z-1)} \\ \frac{1}{z(z-1)} \\ \frac{(z-2)}{z(z-1)} \end{bmatrix}
\end{aligned}$$

(where we have highlighted the fact that, for the computation of $X_f(z)$, the first two columns of $\operatorname{adj}(zI_3 - F)$ are unnecessary) in order to compute the forced state evolution we need to compute the $\mathcal{Z}$-transform of each of the components of the vector $X_f(z)$. It holds that

$$\begin{aligned}
\frac{1}{z^2(z-1)} &= \delta_{-1}(k-3) \\
\frac{1}{z(z-1)} &= \delta_{-1}(k-2) \\
\frac{(z-2)}{z(z-1)} &= \delta(k-1) - \delta_{-1}(k-2)
\end{aligned}$$

The previous expression can be rewritten as follows:

$$\mathbf{x}_f(k) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \delta(k-1) + \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \delta(k-2) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \delta_{-1}(k-3).$$

♠

Exercise 1.8.4 Let us consider the discrete time state space model described by the following equations:

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + Gu(k) &= \begin{bmatrix} 2 & 0 \\ 1 & -2 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(k), \\ y(k) &= H\mathbf{x}(k) &= [0 \quad 1]\,\mathbf{x}(k), \qquad\qquad k \in \mathbb{Z}_+. \end{aligned}$$

i) Determine the Jordan form of the matrix F, the elementary modes of the system and their (convergent, limited, unlimited) behaviour;

ii) determine the $\mathcal{Z}$-transform of the output signals associated with the following conditions:

- $\mathbf{x}(0) = [\,0 \quad 1\,]^T$ and $u(k) = a^k, k \geq 0$, with $a \in \mathbb{R}$;
- $\mathbf{x}(0) = [\,2 \quad 1\,]^T$ and $u(k) = k\ a^k + 1, k \geq 0$, with $a \in \mathbb{R}$.

iii) Determine the initial conditions that ensure that the unforced state evolution is entirely included in a line passing through the origin.

Exercise 1.8.5 Let $\Sigma = (F, G, H)$ be a discrete time state space dynamical system of dimension n. Answer the following questions, providing an adequate explanation:

i) What is the minimum value of n such that $\binom{k}{1}\sin(k-1)$ and $\binom{k}{1}2^{k-1}$ may appear as modes of Σ?

ii) Provided that the (all and only) distinct modes of the system are 2^k, $\binom{k}{1}\,2^{k-1}$, $\binom{k}{2}\,2^{k-2}$, $(-1)^k$, $\binom{k}{1}\cdot\,(-1)^{k-1}$, what is the maximum possible value for n?

Exercise 1.8.6 Let us consider the discrete time state space model described by the following equations:

$$\begin{aligned} \mathbf{x}(k+1) &= F\mathbf{x}(k) + Gu(k) &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1/2 & -1 \\ 0 & 0 & 1/2 \end{bmatrix} \mathbf{x}(k) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(k), \\ y(k) &= H\mathbf{x}(k) &= [0 \quad 1 \quad 0]\,\mathbf{x}(k), \qquad\qquad k \in \mathbb{Z}_+. \end{aligned}$$

i) Determine the Jordan form of the matrix F, the elementary modes of the system and their (convergent, limited, unlimited) behaviour;

ii) determine the $\mathcal{Z}$-transform of the output signal corresponding to the following initial conditions:

- $\mathbf{x}(0) = [\,0 \quad 1 \quad 0\,]^T$ and $u(k) = \delta(k) - a^k \delta_{-1}(k)$, with $a \in \mathbb{R}$;
- $\mathbf{x}(0) = [\,2 \quad 0 \quad 0\,]^T$ and $u(k) = \Big(k\ 1^k + (-1)^k\Big)\delta_{-1}(k)$.

Example 1.8.7 Let us consider the discrete time linear SISO system $\Sigma = (F, g, H)$. By assuming that corresponding to the input signal

$$u(k) = \begin{cases} 1, & k = 0, \\ b^{k-1}(b-a), & k \geq 1, \end{cases}$$

with a and b real parameters, the output forced evolution of the system is

$$y(k) = y_f(k) = \begin{cases} 0, & k \leq 1, \\ -2, \left(-\frac{1}{2}\right)^{k-1} & k \geq 2, \end{cases}$$

determine for what values of the pair a and b the system Σ is BIBO stable.

SOLUTION The $\mathcal{Z}$-transform of the input signal is

$$\begin{aligned} U(z) &= \sum_{t\,\geq 0} u(t) z^{-t} = 1 + \sum_{t\,\geq 1} b^{t-1}(b-a)z^{-t} = 1 + (b-a)z^{-1} \sum_{t\,\geq 1} b^{t-1} z^{-t+1} \\ &= 1 + (b-a)z^{-1} \sum_{t\,\geq 0} b^t z^{-t} = 1 + \frac{(b-a)z^{-1}}{1 - bz^{-1}} = 1 + \frac{b-a}{z-b} = \frac{z-a}{z-b}. \end{aligned}$$

The $\mathcal{Z}$-transform of the output is given by

$$\begin{aligned} Y(z) &= \sum_{t\,\geq 0} y(t) z^{-t} = -2 \sum_{t\,\geq 2} \left(-\frac{1}{2}\right)^{t-1} z^{-t} = -2z^{-1} \sum_{t\,\geq 2} \left(-\frac{1}{2}\right)^{t-1} z^{-t+1} \\ &= -2z^{-1} \sum_{t\,\geq 1} \left(-\frac{1}{2}\right)^{t} z^{-t} = -2z^{-1} \left(\sum_{t\,\geq 0} \left(-\frac{1}{2}\right)^{t} z^{-t} - 1 \right) \\ &= -2z^{-1} \left(\frac{1}{1 + \frac{z^{-1}}{2}} - 1 \right) = \frac{1}{z(z+1/2)}. \end{aligned}$$

Consequently, the transfer function of the system is

$$W(z) = \frac{Y(z)}{U(z)} = \frac{1}{z(z+1/2)} \cdot \frac{z-b}{z-a} = \frac{z-b}{z(z-a)(z+1/2)}. \tag{1.24}$$

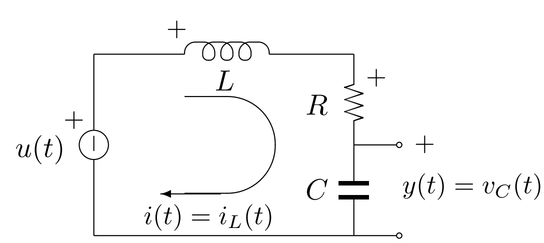
In order to guarantee the BIBO stability of the system it is necessary and sufficient that all the poles of the transfer function have moduli smaller than one; therefore the following cases may occur: if a is a pole of the transfer function (namely the factor $z - a$ does not cancel out with the factor $z - b$, which is to say that a and b are distinct), then $|a| < 1$. If $a = b$, then any value is suitable. To conclude, the system is BIBO stable either if $|a| < 1$ and b is arbitrary or if $|a| \geq 1$, but $b = a$. ♠

Chapter 2

Continuous Time State Space Models

2.1 The RLC Circuit Example

Let us consider the RLC circuit controlled in voltage, with R, L and C positive coefficients denoting the resistance, the inductance and the capacitance, respectively.



The voltage $u(t)$ provided by the voltage generator acts as an input to the system, on the other hand, the voltage measured on the capacitor represents the system output. In the description of the dynamics of a passive electric circuit two variables play a leading role: the current in the inductors and the voltage in the capacitors, namely $i_L(t)$ and $v_C(t)$, respectively. Such quantities, in fact, behave as "memory elements" for the system, by this meaning that in order to determine the values of current and voltage in the circuit starting from a certain time instant t_0 it is necessary to know not only the values of current and voltage provided by the generators at the time instant t_0, but also the values of such variables at the initial time instant. Therefore let us define

$$x_1(t) \doteq i_L(t), \qquad x_2(t) \doteq v_C(t).$$

Basic knowledge about Electric Circuits Theory allows us to determine the differential equations describing the circuit dynamics. The voltage balancing on the only mesh of the circuit leads to

$$u(t) = v_L(t) + v_R(t) + v_C(t),$$

where $v_L(t)$ and $v_R(t)$ represent the voltage drop at the capacitor and inductor heads, respectively. Moreover, by exploiting the relation between current and voltage of the resistor, inductor and capacitor and in the light of the fact that in this specific case the current is the same for all the components, that is $x_1(t)$, we obtain the following equations:

$$\begin{aligned} x_1(t) &= C\,\frac{dv_C(t)}{dt} = C\,\frac{dx_2(t)}{dt}, \\ v_L(t) &= L\,\frac{dx_1(t)}{dt}, \\ v_R(t) &= R\,x_1(t). \end{aligned}$$

By merging together these three identities and the previous equation we obtain

$$\begin{aligned} u(t) &= L\,\frac{dx_1(t)}{dt} + Rx_1(t) + x_2(t), \\ x_1(t) &= C\,\frac{dx_2(t)}{dt}, \end{aligned}$$

from which it follows

$$\begin{aligned} \frac{dx_1(t)}{dt} &= -\frac{R}{L}\,x_1(t) - \frac{1}{L}\,x_2(t) + \frac{1}{L}\,u(t), \\ \frac{dx_2(t)}{dt} &= \frac{1}{C}\,x_1(t). \end{aligned}$$

The output can be easily deduced once we notice that $y(t) = v_C(t)$ and therefore it holds that

$$y(t) = x_2(t).$$

The overall model, expressed in matrix form, is

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u(t), \\ y(t) &= [0 \quad 1]\,\mathbf{x}(t). \end{aligned}$$

2.2 Continuous Time State Space Models

A generic continuous time LTI state space model of **dimension** n, with m **inputs** and p **outputs**, is described by a system of (first order) differential equations of the type

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) + G\mathbf{u}(t), \tag{2.1}$$

$$\mathbf{y}(t) = H\mathbf{x}(t) + D\mathbf{u}(t), \tag{2.2}$$

where $\mathbf{x}$ is the n-dimensional state vector, $\mathbf{u}$ is the m-dimensional input vector and $\mathbf{y}$ is the p-dimensional output vector. The matrices $F \in \mathbb{R}^{n\times n}$, $G \in \mathbb{R}^{n\times m}$, $H \in \mathbb{R}^{p\times n}$ and $D \in \mathbb{R}^{p\times m}$ are arbitrary. Equation (2.1) is known as **state equation**, while (2.2) is the **output equation**. The matrix F is the **state matrix**.

Such a state space model represents an LTI causal system and it is usually assumed that the system dynamics is observed starting from a certain time instant t_0 (without loss of generality it is assumed $t_0 = 0$). This state space model (2.1)-(2.2) is said to be **proper** since in the output equation the input appears while its derivatives do not (this is reasonable from a physical perspective). In particular, if $D = 0$ and hence the output $\mathbf{y}(t)$ does not instantaneously depend on the input $\mathbf{u}(t)$, the system is said to be **strictly proper**.

The dynamical representation of a continuous time state space model functioning can be shown through a block diagram, where the symbol $\int$ represents the integrator block.

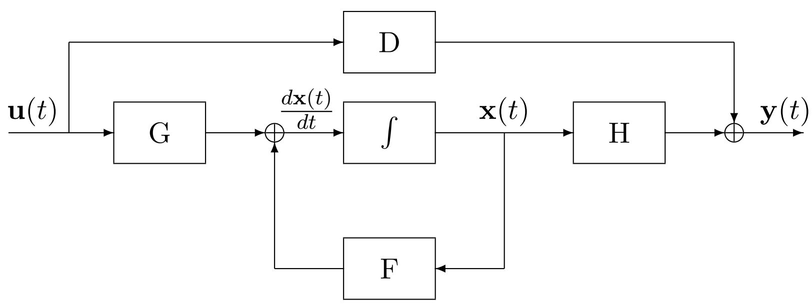


Figure 2.1: *Block diagram of a continuous time state space model.*

In order to uniquely determine the state and output evolution at each time instant $t \geq 0$ it is necessary and sufficient to know the initial state $\mathbf{x}(0)$ starting from which the system evolves[1] at the time instant $t = 0$ and the control input $\mathbf{u}(t)$ for $t \geq 0$. Initial state and input (for nonnegative time instants) represent the two causes acting on the system, while the state evolution $\mathbf{x}(t)$ and output evolution $\mathbf{y}(t)$, for $t \in \mathbb{R}_+$, represent the corresponding effects. Since the system is linear, the superposition principle holds, and by exploiting the same reasoning as in Chapter 1 for discrete time systems, we can decompose the state and output evolution for $t \geq 0$, in an unforced evolution component and in a forced evolution component. In other words, we can express $\mathbf{x}(t)$ and $\mathbf{y}(t)$ in the form

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{x}_\ell(t) + \mathbf{x}_f(t),\\ \mathbf{y}(t) &= \mathbf{y}_\ell(t) + \mathbf{y}_f(t), \quad t \in \mathbb{R}_+,\end{aligned}$$

[1]It is worth noticing that, in order to take into account impulse inputs applied in the origin, the initial time instant value is $t = 0^-$. However, in Systems Theory, the initial condition is usually denoted by the symbol $\mathbf{x}(0)$. It is agreed that, when necessary, $\mathbf{x}(0)$ is meant to be $\mathbf{x}(0^-)$.

where $\mathbf{x}_\ell$ and $\mathbf{y}_\ell$ represent the **unforced state and output evolution** components, respectively, while $\mathbf{x}_f$ and $\mathbf{y}_f$ represent the **forced state and output evolution** components, respectively.

In the following we will address the problem of determining the explicit expression for each of the aforementioned components.

2.3 Unforced Evolution Dynamics

Studying the unforced evolution of the system (2.1)÷(2.2) corresponding to a generic initial condition $\mathbf{x}(0) = \mathbf{x}_0$ is equivalent to the study of the evolution of the **autonomous** system (namely the system without any input applied to it):

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= F\mathbf{x}(t), && (2.3)\\ \mathbf{y}(t) &= H\mathbf{x}(t), \qquad t \in \mathbb{R}_+, && (2.4)\\ \mathbf{x}(0) &= \mathbf{x}_0 \ \in \mathbb{R}^n. && (2.5) \end{aligned}$$

For the study of the unforced evolution it is convenient to first analyse a simple case. Let us consider a system of unitary dimension, namely whose dimension n of the vector state is equal to 1, thus the matrix F collapses into a scalar. In this case, as known from the Mathematical courses, the state expression of the autonomous system (2.3)÷(2.5) at the generic time instant $t \in \mathbb{R}_+$ is

$$\mathbf{x}(t) = e^{Ft}\mathbf{x}_0,$$

and, consequently, the output is

$$\mathbf{y}(t) = He^{Ft}\mathbf{x}_0.$$

We want now to prove that such expressions hold also when n is a generic positive integer, provided that we give to $e^{Ft}, t \in \mathbb{R}_+$ an appropriate meaning when F is a matrix.

2.3.1 Exponential of a Matrix

Definition 2.3.1 *Let $F \in \mathbb{C}^{n\times n}$ be a generic complex matrix. We define the* **exponential of the matrix** *F, denoted by the expression e^{Ft} or, equivalently, by the expression $\exp(Ft)$, the function taking values in $\mathbb{C}^{n\times n}$ defined, for every $t \in \mathbb{R}$, by the following series:*

$$e^{Ft} \doteq I_n + F\ t + F^2\ \frac{t^2}{2!} + F^3\ \frac{t^3}{3!} + \ldots = \sum_{i=0}^{+\infty} F^i\ \frac{t^i}{i!}. \qquad (2.6)$$

Let us evaluate, now, the exponential of some simple matrices.

Example 2.3.2 Let us consider the matrix F of the Example 1.4.1

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

In order to evaluate the expression of the exponential of F we exploit the expression of the generic i-th power of the matrix F, previously determined:

$$F^i = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (-1)^i & 0 \\ 0 & 0 & 2^i \end{bmatrix}, \qquad \forall\, i \in \mathbb{Z}_+.$$

By exploiting the definition of exponential matrix, we get

$$\begin{aligned} e^{Ft} &= \sum_{i=0}^{+\infty} \begin{bmatrix} 1 & 0 & 0 \\ 0 & (-1)^i & 0 \\ 0 & 0 & 2^i \end{bmatrix} \frac{t^i}{i!} = \begin{bmatrix} \displaystyle\sum_{i=0}^{+\infty} 1 \cdot \frac{t^i}{i!} & 0 & 0 \\ 0 & \displaystyle\sum_{i=0}^{+\infty} (-1)^i \cdot \frac{t^i}{i!} & 0 \\ 0 & 0 & \displaystyle\sum_{i=0}^{+\infty} 2^i \cdot \frac{t^i}{i!} \end{bmatrix} \\ &= \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{2t} \end{bmatrix}. \end{aligned}$$

It is immediate to notice that if the matrix F is a diagonal matrix, then its exponential is a diagonal matrix whose (i,i)-th element coincides with the exponential of the (i,i)-th element of F. ♠

Example 2.3.3 Let us consider the matrix F of the Example 1.4.2

$$F = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 3},$$

whose generic i-th power is given by

$$F^i = \begin{bmatrix} 1 & -i & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3^i \end{bmatrix}, \qquad \forall\, i \in \mathbb{Z}_+.$$

It is immediate to notice that

$$
\begin{aligned}
e^{Ft} &= \sum_{i=0}^{+\infty} \begin{bmatrix} 1 & -i & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3^i \end{bmatrix} \frac{t^i}{i!} = \begin{bmatrix} \sum_{i=0}^{+\infty} 1 \cdot \frac{t^i}{i!} & -\sum_{i=0}^{+\infty} i \cdot \frac{t^i}{i!} & 0 \\ 0 & \sum_{i=0}^{+\infty} 1 \cdot \frac{t^i}{i!} & 0 \\ 0 & 0 & \sum_{i=0}^{+\infty} 3^i \cdot \frac{t^i}{i!} \end{bmatrix} \\
&= \begin{bmatrix} e^t & -te^t & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^{3t} \end{bmatrix},
\end{aligned}
$$

where we have exploited the fact that

$$
\sum_{i=0}^{+\infty} i \cdot \frac{t^i}{i!} = \sum_{i=1}^{+\infty} i \cdot \frac{t^i}{i!} = \sum_{i=1}^{+\infty} \frac{t^i}{(i-1)!} = t \cdot \sum_{i=1}^{+\infty} \frac{t^{i-1}}{(i-1)!} = t \cdot \sum_{k=0}^{+\infty} \frac{t^k}{k!} = t \cdot e^t.
$$

♠

Let us focus now on the properties of a given exponential matrix. Then we will show that the aforementioned unforced state and output evolution expressions obtained for the scalar case generalize to the case $n > 1$, provided that we regard e^{Ft} as the exponential of the matrix F. It is worth noticing that among the properties listed in the sequel only the first two are relevant to the aim of determining the expression of $\mathbf{x}_\ell(t)$ and $\mathbf{y}_\ell(t)$. The remaining properties, instead, cover a leading role in the derivation of the explicit formulation of the "elementary modes" of the system as functions of the Jordan form of the system matrix F, as it will be commented upon in section 2.3.3.

2.3.2 Exponential Matrix Properties

In the following we will show the main properties of the exponential function.

1) Value of the exponential function at $t = 0$:

$$
e^{Ft}\big|_{t=0} = I_n.
$$

In fact,

$$
e^{Ft}\big|_{t=0} = \left(\sum_{i=0}^{+\infty} F^i \frac{t^i}{i!} \right)\Bigg|_{t=0} = \left(I_n + F\,t + F^2 \frac{t^2}{2!} + \dots \right)\Bigg|_{t=0} = I_n.
$$

2)Derivative property:

$$\frac{de^{Ft}}{dt} = F \cdot e^{Ft} = e^{Ft} \cdot F.$$

In fact,

$$\frac{de^{Ft}}{dt} = \frac{d}{dt}\left(\sum_{i=0}^{+\infty} F^i \ \frac{t^i}{i!}\right) = \sum_{i=1}^{+\infty} F^i \ i\frac{t^{i-1}}{i!} = F \cdot \sum_{i=1}^{+\infty} F^{i-1} \ \frac{t^{i-1}}{(i-1)!} = Fe^{Ft}.$$

The commutativity of the product between F and e^{Ft} is obvious.

3) Invertibility of the exponential matrix:

For any matrix $F \in \mathbb{C}^{n\times n}$ the inverse of the exponential matrix e^{Ft} exists and coincides with $e^{(-F)t}$, namely the exponential of the matrix $-F$. In fact, it is sufficient to notice that

$$\begin{aligned} e^{Ft} \cdot e^{(-F)t} &= \left(I_n + F\ t + F^2\ \frac{t^2}{2!} + F^3\ \frac{t^3}{3!} + \dots\right)\left(I_n - F\ t + F^2\ \frac{t^2}{2!} - F^3\ \frac{t^3}{3!} + \dots\right) \\ &= I_n \end{aligned}$$

and also that

$$\begin{aligned} e^{(-F)t} \cdot e^{Ft} &= \left(I_n - F\ t + F^2\ \frac{t^2}{2!} - F^3\ \frac{t^3}{3!} + \dots\right)\left(I_n + F\ t + F^2\ \frac{t^2}{2!} + F^3\ \frac{t^3}{3!} + \dots\right) \\ &= I_n. \end{aligned}$$

4) Transpose property:

$$(e^{Ft})^T = e^{F^T t}.$$

In fact, by exploiting the identity $(F^i)^T = (F^T)^i$, $i \in \mathbb{Z}_+$, we get

$$(e^{Ft})^T = \left(\sum_{i=0}^{+\infty} F^i\ \frac{t^i}{i!}\right)^T = \sum_{i=0}^{+\infty} (F^i)^T\ \frac{t^i}{i!} = \sum_{i=0}^{+\infty} (F^T)^i\ \frac{t^i}{i!} = e^{F^T t}.$$

5) Exponential of a scalar matrix:

Let $F \in \mathbb{C}^{n\times n}$ be a scalar matrix, by this meaning a diagonal matrix that can be expresses in the form $F = \lambda I_n$ for some $\lambda \in \mathbb{C}$. Then, the exponential of the matrix F is a scalar matrix as well, specifically,

$$e^{Ft} = e^{\lambda t} I_n = \begin{bmatrix} e^{\lambda t} & & & \\ & e^{\lambda t} & & \\ & & \ddots & \\ & & & e^{\lambda t} \end{bmatrix}.$$

In fact it is immediate to notice that $F^i = \lambda^i I_n$ therefore

$$e^{Ft} = \sum_{i=0}^{+\infty} (\lambda I_n)^i\ \frac{t^i}{i!} = \sum_{i=0}^{+\infty} \lambda^i\ I_n \frac{t^i}{i!} = \left(\sum_{i=0}^{+\infty} \lambda^i\ \frac{t^i}{i!}\right) I_n = e^{\lambda t} I_n.$$

6) Product of the exponential of two commuting matrices:

Let $F_1, F_2 \in \mathbb{C}^{n\times n}$ be two square matrices (of the same dimensions) whose product is commutative, namely $F_1F_2 = F_2F_1$. Then their exponential commute as well and it holds

$$e^{F_1t} \cdot e^{F_2t} = e^{(F_1+F_2)t} = e^{F_2t} \cdot e^{F_1t}.$$

Vice versa, if F_1 e F_2 do not commute, the three above expressions are all distinct.

In fact

$$\begin{aligned} e^{F_1t} \cdot e^{F_2t} &= \left(\sum_{i=0}^{+\infty} F_1^i \, \frac{t^i}{i!}\right)\left(\sum_{i=0}^{+\infty} F_2^i \, \frac{t^i}{i!}\right) \\ &= I_n + (F_1+F_2)t + (F_1^2 + 2F_1F_2 + F_2^2)\, \frac{t^2}{2!} + \dots \end{aligned}$$

and

$$\begin{aligned} e^{F_2t} \cdot e^{F_1t} &= \left(\sum_{i=0}^{+\infty} F_2^i \, \frac{t^i}{i!}\right)\left(\sum_{i=0}^{+\infty} F_1^i \, \frac{t^i}{i!}\right) \\ &= I_n + (F_1+F_2)t + (F_1^2 + 2F_2F_1 + F_2^2)\, \frac{t^2}{2!} + \dots \end{aligned}$$

while

$$\begin{aligned} e^{(F_1+F_2)t} &= \sum_{i=0}^{+\infty} (F_1+F_2)^i \, \frac{t^i}{i!} = I_n + (F_1+F_2)t + (F_1+F_2)^2 \, \frac{t^2}{2!} + \dots \\ &= I_n + (F_1+F_2)t + (F_1^2 + F_1F_2 + F_2F_1 + F_2^2)\, \frac{t^2}{2!} + \dots . \end{aligned}$$

Then it is immediate to verify that the three expressions coincide if and only if $F_1F_2 = F_2F_1$. In this case, in fact, the matrices behave as scalars and consequently also their exponentials behave as exponentials of scalar values.

7) Exponential of a Jordan miniblock:

Let us consider the Jordan miniblock of dimension ν associated with the complex number λ, i.e. the matrix

$$J_\lambda = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{bmatrix} \in \mathbb{C}^{\nu\times\nu}. \tag{2.7}$$

The expression of J_λ is

$$e^{J_\lambda t} = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2!}\, e^{\lambda t} & \dots & \frac{t^{\nu-1}}{(\nu-1)!}\, e^{\lambda t} \\ & e^{\lambda t} & te^{\lambda t} & \ddots & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & e^{\lambda t} & t\, e^{\lambda t} \\ & & & & e^{\lambda t} \end{bmatrix}. \tag{2.8}$$

In particular, for $\lambda = 0$, one gets

$$e^{J_0 t} = \begin{bmatrix} 1 & t & \frac{t^2}{2!} & \dots & \frac{t^{\nu-1}}{(\nu-1)!} \\ & 1 & t & \ddots & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & 1 & t \\ & & & & 1 \end{bmatrix}. \tag{2.9}$$

As already mentioned in Chapter 1, the matrix J_0 has a finite number of non null powers and, specifically,

$$(J_0)^2 = \begin{bmatrix} 0 & 0 & 1 & \dots & 0 \\ & 0 & 0 & 1 & \vdots \\ & & \ddots & \ddots & 1 \\ & & & 0 & 0 \\ & & & & 0 \end{bmatrix}, \qquad (J_0)^3 = \begin{bmatrix} 0 & 0 & 0 & 1 & \dots & & \dots & 0 \\ & 0 & 0 & 0 & 1 & & & \vdots \\ & & \ddots & \ddots & & \ddots & & \vdots \\ & & & \ddots & \ddots & & \ddots & \vdots \\ & & & & \ddots & \ddots & & 1 \\ & & & & & 0 & 0 & 0 \\ & & & & & & 0 & 0 \\ & & & & & & & 0 \end{bmatrix},$$

$$(J_0)^{\nu-1} = \begin{bmatrix} 0 & 0 & 0 & \dots & 1 \\ & 0 & 0 & 0 & \vdots \\ & & \ddots & \ddots & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{bmatrix} \qquad (J_0)^\nu = 0.$$

Consequently

$$e^{J_0 t} = \sum_{i=0}^{\nu-1} (J_0)^i \, \frac{t^i}{i!}$$

has the same expression as in (2.9). Since the following identity holds

$$J_\lambda = \lambda I_\nu + J_0,$$

λI_ν is a scalar matrix, and λI_ν and J_0 commute, then by exploiting the properties 5) and 6) one gets

$$e^{J_\lambda t} = e^{(\lambda I_\nu + J_0)t} = e^{(\lambda I_\nu)t} \cdot e^{J_0 t} = e^{\lambda t} I_\nu \cdot e^{J_0 t} = e^{\lambda t} e^{J_0 t}.$$

The result follows from the first part of the proof (case $\lambda = 0$).

8) Exponential of a block diagonal matrix:

Let F be a block diagonal matrix, namely a matrix having a structure of the type

$$F = \begin{bmatrix} F_1 & & & \\ & F_2 & & \\ & & \ddots & \\ & & & F_r \end{bmatrix},$$

where F_i is a square matrix of dimension $n_i \times n_i$, then

$$e^{Ft} = \begin{bmatrix} e^{F_1 t} & & & \\ & e^{F_2 t} & & \\ & & \ddots & \\ & & & e^{F_r t} \end{bmatrix}.$$

The proof trivially follows from the fact that the i-th power of the matrix F is

$$F^i = \begin{bmatrix} (F_1)^i & & & \\ & (F_2)^i & & \\ & & \ddots & \\ & & & (F_r)^i \end{bmatrix}, \quad \forall\, i \in \mathbb{Z}_+.$$

9) Similarity relation:

Let F_1 and F_2 be two similar matrices in $\mathbb{C}^{n\times n}$, by this meaning that there exists a non singular matrix $T \in \mathbb{C}^{n\times n}$ such that $F_2 = T^{-1} F_1 T$. Then the exponentials of the two matrices are related by the following expression:

$$e^{F_2 t} = T^{-1} e^{F_1 t} T.$$

In fact,

$$\begin{aligned} e^{F_2 t} &= \sum_{i=0}^{+\infty} F_2^i \, \frac{t^i}{i!} = \sum_{i=0}^{+\infty} (T^{-1} F_1 T)^i \, \frac{t^i}{i!} = \sum_{i=0}^{+\infty} T^{-1} F_1^i T \, \frac{t^i}{i!} \\ &= T^{-1} \left(\sum_{i=0}^{+\infty} F_1^i \, \frac{t^i}{i!} \right) T = T^{-1} e^{F_1 t} T. \end{aligned}$$

10) Eigenvectors and eigenvalues of an exponential matrix:

Let $\mathbf{v}$ be an **eigenvector** of the matrix $F \in \mathbb{C}^{n\times n}$, corresponding to the **eigenvalue** λ. Then for every fixed $t \in \mathbb{R}$, $\mathbf{v}$ is eigenvector of the matrix e^{Ft} related to the eigenvalue $e^{\lambda t}$.

In fact, by exploiting the fact that $F^i\mathbf{v} = \lambda^i\mathbf{v}$, for every $i \in \mathbb{Z}_+$, one gets

$$e^{Ft}\mathbf{v} = \left(\sum_{i=0}^{+\infty} F^i\ \frac{t^i}{i!}\right)\mathbf{v} = \sum_{i=0}^{+\infty} (F^i\mathbf{v})\ \frac{t^i}{i!} = \sum_{i=0}^{+\infty} (\lambda^i\mathbf{v})\ \frac{t^i}{i!} = \left(\sum_{i=0}^{+\infty} \lambda^i\ \frac{t^i}{i!}\right)\mathbf{v} = e^{\lambda t}\mathbf{v}.$$

2.3.3 Elementary Modes and Unforced Evolution Expression

Once we have introduced the concept of exponential matrix $e^{Ft}, t \in \mathbb{R}$, and its properties, we are now in a position to derive the expression of the unforced evolution of a state space model.

Proposition 2.3.4 *The state and output evolution of the system (2.1)÷(2.2), starting from an arbitrary initial condition* $\mathbf{x}(0) = \mathbf{x}_0$ *and in the absence of input signal, are*

$$\mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}_0, \qquad t \in \mathbb{R}_+, \tag{2.10}$$

and

$$\mathbf{y}_\ell(t) = He^{Ft}\mathbf{x}_0, \qquad t \in \mathbb{R}_+, \tag{2.11}$$

respectively.

Proof One needs to demonstrate that the system

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= F\mathbf{x}(t), \qquad t \in \mathbb{R}_+, \\ \mathbf{x}(0) &= \mathbf{x}_0, \end{aligned}$$

with F a real matrix of dimension $n \times n$, has solution

$$\mathbf{x}(t) = \mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}_0, \qquad t \in \mathbb{R}_+.$$

This holds true due to the derivative property

$$\frac{d\mathbf{x}(t)}{dt} = \frac{d}{dt}\left(e^{Ft}\mathbf{x}_0\right) = Fe^{Ft}\mathbf{x}_0 = F\mathbf{x}(t),$$

(and this proves that the given expression is a solution of the differential equation). Moreover, in the light of property 1), one gets

$$\mathbf{x}(0) = e^{Ft}\mathbf{x}_0\big|_{t=0} = I_n\mathbf{x}_0 = \mathbf{x}_0.$$

As far as the output expression is concerned, it holds that

$$\mathbf{y}_\ell(t) = H\mathbf{x}_\ell(t), \qquad t \in \mathbb{R}_+,$$

therefore

$$\mathbf{y}_\ell(t) = He^{Ft}\mathbf{x}_0, \qquad t \in \mathbb{R}_+.$$

■

At this point, one may wonder how to effectively evaluate the exponential of a given matrix F. To this end we exploit the aforementioned properties of the exponential of a matrix and the matrix Jordan form. In fact, it is immediate to notice that, since the computation of the exponential of a matrix in Jordan form can be easily obtained (in light of properties 7) and 8)) and since every matrix is similar to a matrix in Jordan form, by exploiting property 9) one gets the following results.

Proposition 2.3.5 *Consider the matrix $F \in \mathbb{C}^{n\times n}$ with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_r$, and let $T \in \mathbb{C}^{n\times n}$ be a non singular matrix such that $T^{-1}FT$ is in Jordan form:*

$$J \doteq T^{-1}FT = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_r \end{bmatrix},$$

where

$$J_i = \begin{bmatrix} J_{i,1} & & & \\ & J_{i,2} & & \\ & & \ddots & \\ & & & J_{1,s_i} \end{bmatrix},$$

and $J_{i,k}$ is the k-th Jordan miniblock of dimension $n_{i,k}$, corresponding to the eigenvalue λ_i. Then,

$$e^{Ft} = Te^{Jt}T^{-1} = T \begin{bmatrix} e^{J_1 t} & & & \\ & e^{J_2 t} & & \\ & & \ddots & \\ & & & e^{J_r t} \end{bmatrix} T^{-1},$$

where every matrix $e^{J_i t}$ is a block diagonal matrix, whose k-th diagonal block is

$$e^{J_{i,k}t} = \begin{bmatrix} e^{\lambda_i t} & te^{\lambda_i t} & \frac{t^2}{2!}\, e^{\lambda_i t} & \dots & \frac{t^{n_{i,k}-1}}{(n_{i,k}-1)!}\, e^{\lambda_i t} \\ & e^{\lambda_i t} & te^{\lambda_i t} & \ddots & \vdots \\ & & \ddots & \ddots & \vdots \\ & & & e^{\lambda_i t} & t\, e^{\lambda_i t} \\ & & & & e^{\lambda_i t} \end{bmatrix}. \tag{2.12}$$

In the following, for the sake of simplicity, we will assume that the miniblocks corresponding to the same eigenvalue are arranged in decreasing order of size.

Each or the elementary (possibly complex) functions $\frac{t^k}{k!}\ e^{\lambda_i t}$ is called an **elementary mode** of the matrix F or, equivalently, of the system. Notice that, as it happened for the input/output models studied in the Automatic Control course, all the elementary modes involved in the expression of the unforced evolution are exponential functions, possibly multiplied by a polynomial function in the variable t. Provided that the unforced output evolution of a state space model can be expressed in the form $\mathbf{y}(t) = He^{Ft}\mathbf{x}_0$, from the aforementioned proposition it turns out that each state and output component, $x_j(t)$ and $y_j(t)$, in unforced evolution, is the linear combination of the elementary modes of the system. The combinatorial coefficients depend on the matrix T (and on its inverse) and on the initial conditions vector. As already mentioned for discrete time state space models, not all the elementary modes appearing in the expression of the unforced state evolution appear in the unforced output evolution, due to the presence of the matrix H. In general, in fact, only a subset of the elementary modes is "observable", namely detectable form the unforced output evolution, interpreted as "observation of the system".

If F is a real matrix, the possibly complex eigenvalues appear in conjugate pairs and, as already mentioned for the discrete time models, if in the Jordan form of F there exists a Jordan miniblock of dimension ν corresponding to the complex eigenvalue $\sigma + j\omega$ (with $\sigma, \omega \in \mathbb{R}$), then there exists a Jordan miniblock of dimension ν corresponding to the conjugate eigenvalue $\sigma - j\omega$. Consequently, if F is a real matrix, its complex modes appear in conjugate pairs and are combined by complex coefficients so that the final expression is real. Therefore it is always possible to replace the pair of complex modes $e^{(\sigma+j\omega)t}, e^{(\sigma-j\omega)t}$ with the pair of real modes $e^{\sigma t}\cos(\omega t)$, $e^{\sigma t}\sin(\omega t)$. The same holds true for complex modes of the type $\frac{t^k}{k!}e^{(\sigma+j\omega)t}$, $\frac{t^k}{k!}e^{(\sigma-j\omega)t}$.

As regards the distinct modes involved in the unforced state evolution, as $\mathbf{x}_0 \in \mathbb{R}^n$ varies, by exploiting a reasoning analogous to the one already adopted for discrete time models, we can say that the maximum number of distinct elementary modes associated with the eigenvalue λ_i is equal to the dimension of the largest Jordan miniblock corresponding to λ_i. Therefore, if we assume that the Jordan miniblocks are ordered by decreasing size, it coincides with $n_{i,1}$. So, the total number of distinct elementary modes of the F matrix is equal to $\sum_{i=1}^r n_{i,1}$ and they can be uniquely determined by the minimal polynomial of F, $\Psi_F(s)$. We will remark this concept later on, when dealing with the concept of state space models stability.

Example 2.3.6 Let us consider the matrix F of the Example 2.3.3:

$$F = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 3},$$

for which we have already evaluated the expression of the generic power F^i thus deriving its exponential, namely:

$$e^{Ft} = \begin{bmatrix} e^t & -te^t & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^{3t} \end{bmatrix}.$$

The matrix F is not in Jordan form, but its structure is very close to the Jordan one. Let us see, now, how the expression of e^{Ft} can be derived starting from the aforementioned reasoning referring to the Jordan form. As a first step we observe that F has two distinct eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 3$, with algebraic multiplicities $\mu_1 = 2$ and $\mu_2 = 1$, respectively. A quick calculation shows that the dimension of the eigenspace of F (and hence of the eigenspace of its Jordan form as well) corresponding to the eigenvalue λ_1 is equal to 1, namely the geometric multiplicity of λ_1 is 1. This allows us to deduce that the Jordan form of F is

$$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

The matrix T that relates F to J is determined by imposing $FT = TJ$ (and, obviously, $\det T \neq 0$), from which it follows that a possible basis change matrix is

$$T = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore for every initial condition $\mathbf{x}_0$ the unforced state evolution can be expressed as

$$\mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}_0 = \begin{bmatrix} e^t & -te^t & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^{3t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix},$$

or, equivalently, as

$$\mathbf{x}_\ell(t) = Te^{Jt}T^{-1}\mathbf{x}_0 = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & te^t & 0 \\ 0 & e^t & 0 \\ 0 & 0 & e^{3t} \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}.$$

♠

Example 2.3.7 Let us consider the matrix

$$F = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \in \mathbb{R}^{2\times 2}.$$

In this case the calculation of the generic power of the matrix F is challenging, while the calculation of its Jordan form is not. The characteristic polynomial of the matrix F is

$$\Delta_F(s) = \det(sI_2 - F) = (s-2)(s-3) - 2 = s^2 - 5s + 4 = (s-1)(s-4),$$

thus the eigenvalues of the matrix are $\lambda_1 = 1$ and $\lambda_2 = 4$ and both are simple eigenvalues. It follows that the Jordan form of F is

$$J = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}.$$

Since the matrix F is diagonalizable, every matrix T such that $T^{-1}FT = J$, is made up of columns that are the eigenvectors corresponding to the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 4$ and arranged in the same order with which the eigenvalues appear in the diagonal matrix J. A possible choice for T is

$$T = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}.$$

It follows that for any initial state $\mathbf{x}_0 \in \mathbb{R}^2$ one gets

$$\mathbf{x}_\ell(t) = Te^{Jt}T^{-1}\mathbf{x}_0 = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{4t} \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}.$$

♠

2.4 Forced Evolution

Let us consider the continuous time LTI state space model

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= F\mathbf{x}(t) + G\mathbf{u}(t), && (2.13) \\ \mathbf{y}(t) &= H\mathbf{x}(t) + D\mathbf{u}(t), \qquad t \in \mathbb{R}_+, && (2.14) \end{aligned}$$

where the variables, the dimensions and the assumptions are the same as before. We want now to determine the state and output forced evolutions, namely the system state and output responses to an input applied when the system is at rest (i.e. $\mathbf{x}(0) = 0$).

Proposition 2.4.1 *The state and output evolutions of the system (2.13)÷(2.14) corresponding to the input signal $\mathbf{u}(t), t \in \mathbb{R}_+$, having no impulsive components, and starting from zero initial conditions are*

$$\mathbf{x}_f(t) = \int_0^t e^{F(t-\tau)}G\mathbf{u}(\tau)d\tau, \qquad t \in \mathbb{R}_+, \tag{2.15}$$

and

$$\mathbf{y}_f(t) = \int_0^t He^{F(t-\tau)}G\mathbf{u}(\tau)d\tau + D\mathbf{u}(t), \qquad t \in \mathbb{R}_+, \tag{2.16}$$

respectively.

PROOF Let us focus on the state expression, since once we have proved it, the identity (2.15) trivially follows from it. From (2.13) one gets

$$\frac{d\mathbf{x}(t)}{dt} - F\mathbf{x}(t) = G\mathbf{u}(t).$$

By the invertibility of the exponential matrix, such equation is solvable for $t \in \mathbb{R}_+$, starting from zero initial conditions, if and only if the following differential equation:

$$e^{-Ft}\left(\frac{d\mathbf{x}(t)}{dt} - F\mathbf{x}(t)\right) = e^{-Ft}G\mathbf{u}(t).$$

is solvable. It is immediate to notice (by exploiting the aforementioned derivative properties) that

$$e^{-Ft}\left(\frac{d\mathbf{x}(t)}{dt} - F\ \mathbf{x}(t)\right) = \frac{d}{dt}\left(e^{-Ft}\mathbf{x}(t)\right).$$

Therefore it holds that

$$\frac{d}{dt}\left(e^{-Ft}\mathbf{x}(t)\right) = e^{-Ft}G\mathbf{u}(t).$$

By integrating both sides of the above equation from 0 to t one gets

$$\int_0^t \frac{d}{d\tau}\left(e^{-F\tau}\mathbf{x}(\tau)\right)d\tau = \int_0^t e^{-F\tau}G\mathbf{u}(\tau)d\tau.$$

It follows that

$$e^{-F\tau}\mathbf{x}(\tau)\Big|_{\tau=0}^{\tau=t} = \int_0^t e^{-F\tau}G\mathbf{u}(\tau)d\tau,$$

namely

$$e^{-Ft}\mathbf{x}(t) - \mathbf{x}(0) = \int_0^t e^{-F\tau}G\mathbf{u}(\tau)d\tau.$$

Keeping in mind that the initial condition $\mathbf{x}(0)$ is zero and by exploiting the invertibility of the exponential matrix and the trivial identity $e^{Ft}\cdot e^{-F\tau} = e^{F(t-\tau)}$, one gets

$$\mathbf{x}(t) = e^{Ft}\int_0^t e^{-F\tau}G\mathbf{u}(\tau)d\tau = \int_0^t e^{F(t-\tau)}G\mathbf{u}(\tau)d\tau.$$

■

If the input signal includes impulsive components located at the origin, it is necessary to modify the aforementioned formulas to take them into account. In practice one needs to replace the lower bound of the integral range, 0, with 0^-, in the equations (2.15) and (2.16).

The i-th column of the matrix function

$$W(t) \doteq He^{Ft}G\delta_{-1}(t) + D\delta(t) \tag{2.17}$$

represents the forced output response to the input $\mathbf{u}(t) = \mathbf{e}_i\delta(t)$ ($\mathbf{e}_i$ being the i-th vector of the canonical basis in $\mathbb{R}^m$). From the previous proposition, it follows that the forced output corresponding to such an input is

$$\mathbf{y}_f^{(i)}(t) = \int_0^t He^{F(t-\tau)}G\mathbf{e}_i\delta(\tau)d\tau + D\mathbf{e}_i\delta(t) = He^{Ft}G\mathbf{e}_i + D\mathbf{e}_i\delta(t), \qquad \text{per } t \in \mathbb{R}_+.$$

For this reason, the matrix $W(t), t \in \mathbb{R}_+$, is called **system impulse response** and the output forced evolution, previously determined, can be interpreted as the convolution between the input and the system impulse response, i.e.

$$\mathbf{y}_f(t) = [W * \mathbf{u}](t), \qquad t \in \mathbb{R}_+.$$

To conclude, the overall equations describing the state and output evolutions, starting from a generic initial condition and corresponding to a generic input are:

$$\mathbf{x}(t) \;=\; \mathbf{x}_\ell(t) + \mathbf{x}_f(t) = e^{Ft}\mathbf{x}_0 + \int_0^t e^{F(t-\tau)}G\mathbf{u}(\tau)d\tau, \tag{2.18}$$

$$\mathbf{y}(t) \;=\; \mathbf{y}_\ell(t) + \mathbf{y}_f(t) = He^{Ft}\mathbf{x}_0 + \int_0^t He^{F(t-\tau)}G\mathbf{u}(\tau)d\tau + D\mathbf{u}(t), \tag{2.19}$$

both holding for $t \in \mathbb{R}_+$.

Notice that, by assuming a generic time instant t_0 as initial time instant, the output and state expressions at the generic time instant $t \geq t_0$ are given by

$$\begin{aligned}
\mathbf{x}(t) &= e^{F(t-t_0)}\mathbf{x}(t_0) + \int_{t_0}^t e^{F(t-\tau)}G\mathbf{u}(\tau)d\tau, \\
\mathbf{y}(t) &= He^{F(t-t_0)}\mathbf{x}(t_0) + \int_{t_0}^t He^{F(t-\tau)}G\mathbf{u}(\tau)d\tau + D\mathbf{u}(t).
\end{aligned}$$

2.5 State Space Model Internal Stability

As for discrete time state space models, it is possible to introduce some stability definitions also for continuous time state space models. Some of them refer to the unforced state evolution, and represent specific forms of **internal stability**. The **external or BIBO stability**, instead, refer to the output forced evolution and will be considered later on in this chapter.

Definition 2.5.1 *The system (2.13)÷(2.14) is said to be*

- **asymptotically stable** *if, for any choice of the initial condition* $\mathbf{x}(0)$*, the unforced state evolution* $\mathbf{x}_\ell(t)$ *asymptotically converges to zero, i.e.*

$$\lim_{t\to+\infty} \mathbf{x}_\ell(t) = 0;$$

- **simply stable** *if, for any choice of the initial condition* $\mathbf{x}(0)$, *the unforced state evolution* $\mathbf{x}_\ell(t)$ *is a bounded function, by this meaning that there exists a scalar* $M \in \mathbb{R}_+$ *such that*

$$\|\mathbf{x}_\ell(t)\| \leq M, \qquad \textit{for every } t \in \mathbb{R}_+;$$

- **unstable** *otherwise.*

We are now in a position to state the following proposition.

Proposition2.5.2 *A continuous time state space model described by equations (2.13)÷(2.14) is*

- *asymptotically stable if and only if all the zeros of the characteristic polynomial of the system* $\Delta_F(s)$ *(equivalently, of the minimal polynomial of the system* $\Psi_F(s)$*) have real parts smaller than zero, namely all the eigenvalues of* F *have real parts smaller than zero;*
- *simply stable if and only if all the zeros of the minimal polynomial of the system* $\Psi_F(z)$ *have real part smaller than or equal to* 0 *and every zero with zero real part has unitary multiplicity. Equivalently, all the eigenvalues of* F *have real parts smaller than or equal to* 0 *and the eigenvalues with zero real part are associated, in the Jordan form of* F*, to miniblocks of unitary dimension.*

Example 2.5.3 Let us consider the following state space model:

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}(t), \qquad t \in \mathbb{R}_+.$$

The characteristic polynomial of the matrix F is $\Delta_F(s) = (s-1)(s+1)$. Therefore, F has a real eigenvalue (with real part) less than zero and a real eigenvalue (with real part) greater than zero, thus the system is unstable. ♠

Example 2.5.4 Let us consider the following state space model:

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}(t), \qquad t \in \mathbb{R}_+.$$

The characteristic polynomial of the matrix F is $\Delta_F(s) = s^2$. Therefore F has an eigenvalue in the origin with algebraic multiplicity 2. One could think that the system is unstable, but this is not the case. As F has no eigenvalues with positive real parts, but has eigenvalues with zero real part, it is necessary to evaluate the dimension of the Jordan miniblocks corresponding to the eigenvalues with zero real part. In this case, F can be easily interpreted as a Jordan form with two miniblocks of dimension 1 corresponding to the 0 eigenvalue. But then, from the previous proposition, it follows that the system is simply (but not asymptotically) stable. The explicit expression of the system solution leads to

$$\mathbf{x}(t) = \mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}_0 = I_n\mathbf{x}_0 = \mathbf{x}_0, \qquad t \in \mathbb{R}_+.$$

♠

Example 2.5.5 Let us consider the following state space model:

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{x}(t), \qquad t \in \mathbb{R}_+.$$

The characteristic polynomial of the matrix F is $\Delta_F(s) = s^2(s+1)$. Therefore, F has a real eigenvalue (with real part) smaller than zero and an eigenvalue in the origin with algebraic multiplicity 2. In order to understand if the system is simply stable or unstable, we need to check the dimension(s) of the Jordan miniblock(s) corresponding to the zero eigenvalue. Since the algebraic multiplicity of the zero eigenvalue is 2, the sum of the dimensions of the miniblocks associated with 0 is 2. Therefore two cases may occur: a) the null eigenvalue is associated with two Jordan miniblocks of dimension 1 (and hence the system is simply stable), b) the null eigenvalue is associated with a single Jordan miniblock of dimension 2, and hence the system is unstable. Since the number of miniblocks corresponding to a given eigenvalue λ coincides, as we have already said, with the geometric multiplicity of the eigenvalue, namely with the dimension of the eigenspace $\ker(\lambda I - F)$, case a) corresponds to the case when $\dim(\ker(0 \cdot I_3 - F)) = 2$, while case b) to the case when $\dim(\ker(0 \cdot I_3 - F)) = 1$. By noticing that

$$\ker(0 \cdot I_3 - F) = \ker\left(\begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) = \langle \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \rangle,$$

it is immediate to notice that case b) occurs and the system is unstable. ♠

2.6 System Analysis in the Laplace Transform Domain

In previous courses, we have introduced the Laplace transform and verified that it can be exploited in the study of the dynamics of causal LTI systems described by n-th order linear differential equations with constant coefficient.

We want now to apply the same strategy to continuous time LTI state space models, described as usual, by the equations

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) + G\mathbf{u}(t), \tag{2.20}$$

$$\mathbf{y}(t) = H\mathbf{x}(t) + D\mathbf{u}(t), \qquad t \in \mathbb{R}_+. \tag{2.21}$$

We assume that the system starts at the time $t = 0$ from an arbitrary initial condition $\mathbf{x}(0) = \mathbf{x}_0$ and we assume the input $\mathbf{u}(t), t \in \mathbb{R}_+$, to be known.

Moreover, we assume that the input signal (is causal and) admits (unilateral) Laplace transform, namely all its components admit (unilateral) Laplace transform. Under these assumptions it is possible to demonstrate that the state and output signals admit Laplace transforms, as well. Therefore we can introduce the following symbols:

$$U(s) \doteq \mathcal{L}[\mathbf{u}(t)], \qquad X(s) \doteq \mathcal{L}[\mathbf{x}(t)], \qquad Y(s) \doteq \mathcal{L}[\mathbf{y}(t)].$$

If we apply the Laplace transform to both sides of the equations (2.20) and (2.21), by exploiting the Laplace transform derivative property, we get:

$$\begin{aligned} sX(s) - \mathbf{x}_0 &= FX(s) + GU(s), \\ Y(s) &= HX(s) + DU(s), \end{aligned}$$

from which it immediately follows:

$$\begin{aligned} X(s) &= (sI_n - F)^{-1}\mathbf{x}_0 + (sI_n - F)^{-1}GU(s), && (2.22) \\ Y(s) &= HX(s) + DU(s) \\ &= H(sI_n - F)^{-1}\mathbf{x}_0 + \left(H(sI_n - F)^{-1}G + D\right)U(s). && (2.23) \end{aligned}$$

Notice that both the Laplace transform of the state and the Laplace transform of the output consist of two terms: one in which the initial condition $\mathbf{x}_0$ appears (while the input does not) and one in which the Laplace transform of the input appears (while the initial condition $\mathbf{x}_0$ does not). But then it is immediate to recognize in the first terms the Laplace transforms of the unforced evolution components and in the second ones the Laplace transforms of the forced evolution components. In fact, by indicating with $\mathbf{x}_\ell$ e $\mathbf{y}_\ell$ the state and unforced output evolution components, respectively and with $\mathbf{x}_f$ and $\mathbf{y}_f$ the state and output forced evolution components, respectively, one gets

$$\begin{aligned} X_\ell(s) &\doteq \mathcal{L}[\mathbf{x}_\ell(t)] = (sI_n - F)^{-1}\mathbf{x}_0, \\ Y_\ell(s) &\doteq \mathcal{L}[\mathbf{y}_\ell(t)] = H(sI_n - F)^{-1}\mathbf{x}_0, \\ X_f(s) &\doteq \mathcal{L}[\mathbf{x}_f(t)] = (sI_n - F)^{-1}GU(s), \\ Y_f(s) &\doteq \mathcal{L}[\mathbf{y}_f(t)] = \left(H(sI_n - F)^{-1}G + D\right)U(s). \end{aligned}$$

The rational matrix

$$W(s) \doteq H(sI_n - F)^{-1}G + D \qquad (2.24)$$

is called **transfer matrix** of the state space model. It relates the Laplace transform of the input to the Laplace transform of the forced output and therefore it has dimensions $p \times m$. It is immediate to notice that $W(s)$ is the (unilateral) Laplace transform of the system impulse response, namely

$$W(s) = \mathcal{L}[W(t)].$$

$W(s)$ is a proper rational matrix, sum of a strictly proper rational component, $H(sI_n - F)^{-1}G$, and of a constant term, D. As the transfer matrix is a Laplace transform, it makes sense to search for its region of convergence. Since we deal with causal systems, it is immediate to notice that the region of convergence of $W(s)$ (resulting from the intersection of the regions of convergence of all its components) is $\mathrm{Re}(s) > \mathrm{Re}(p)$, where p is the pole with maximum real part among the poles of $W(s)$ (equivalently, of its components). Finally, following a reasoning similar to the one adopted for discrete time systems, we claim that, in general,

$$\{\text{poles of } W(s)\} \subseteq \{\text{eigenvalues of } F\}.$$

To conclude, we point out the consistency of the adopted nomenclature. A generic state space model is said to be proper and indeed its transfer function is a proper rational matrix; on the other hand a state space model is strictly proper ($D = 0$) if and only if its transfer matrix is a strictly proper rational matrix.

2.7 External (or BIBO) Stability and Frequency Response

Definition 2.7.1 *A continuous time state space model described by the pair of equations (2.20)÷(2.21) is called* **externally or BIBO stable** *if, starting from zero initial conditions, it responds to each input, zero for $t < 0$ and bounded for $t \geq 0$, with a (forced) output that is zero for $t < 0$ and bounded for $t \geq 0$.*

In the following we give a characterization of BIBO stability. The proof is omitted.

Proposition 2.7.2 *Given a continuous time state space model described by the equations (2.20)÷(2.21), the following facts are equivalent:*

i) the state space model is BIBO stable;

ii) the norm of the system impulse response $W(t), t \in \mathbb{R}_+$, is summable, namely

$$\int_0^{+\infty} \|W(t)\| dt < +\infty; \tag{2.25}$$

iii) all components $w_{ij}(t)$ of the system impulse response $W(t), t \in \mathbb{R}_+$, are summable in norm, namely

$$\int_0^{+\infty} |w_{ij}(t)| dt < +\infty, \qquad \forall\, i, j; \tag{2.26}$$

iv) all the elementary modes appearing (i.e., weighted by a nonzero coefficient) in the expression of the impulse response $W(t), t \in \mathbb{R}_+$, are convergent;

v) all poles of the system transfer matrix $W(s)$ have real part smaller than zero (namely the region of convergence of $W(s)$ includes the imaginary axis).

As for discrete time state space models, it is immediate to notice that asymptotic stability implies BIBO stability, while the opposite is not true. This is because of possible cancellations in the calculation of the transfer matrix. In fact, if all the "unstable" zeros of $\Delta_F(s)$ (namely the zeros located in $\mathrm{Re}(s) \geq 0$) are zeros (of the same multiplicity or of higher multiplicity) of all the elements of $H\mathrm{adj}(sI_n - F)G$, then the system is BIBO stable even it is not asymptotically stable.

2.8 The RLC Circuit Example (contd.)

Let us go back to the RLC circuit example, that we exploited to introduce continuous time state space models. Let us focus now on the dynamics of this model. The characteristic polynomial of the matrix F of the system is

$$\Delta_F(s) = \det \begin{bmatrix} s + \frac{R}{L} & \frac{1}{L} \\ -\frac{1}{C} & s \end{bmatrix} = s^2 + \frac{R}{L}s + \frac{1}{LC}.$$

Depending on the value taken by the discriminant

$$\Delta \doteq \frac{R^2}{L^2} - \frac{4}{LC}$$

the following cases may occur:

- $\Delta > 0$;
- $\Delta = 0$;
- $\Delta < 0$.

For $\Delta > 0$ one gets two distinct real roots,

$$\lambda_{1,2} \doteq -\frac{R}{2L} \pm \frac{1}{2}\sqrt{\frac{R^2}{L^2} - \frac{4}{LC}},$$

and hence two distinct exponential real modes, namely

$$e^{\lambda_1 t}, \qquad e^{\lambda_2 t}, \qquad t \in \mathbb{R}.$$

For $\Delta = 0$ one gets a real root with algebraic multiplicity 2,

$$\lambda \doteq -\frac{R}{2L}.$$

To understand if such a root is associated with just one mode or with two modes, we need to verify the geometric multiplicity of the eigenvalue λ. To this end we compute

$$\dim\ (\ker(\lambda I_2 - F))$$

or, equivalently,

$$\operatorname{rank}(\lambda I_2 - F) = \operatorname{rank}\left(\begin{bmatrix} \lambda + \frac{R}{L} & \frac{1}{L} \\ -\frac{1}{C} & \lambda \end{bmatrix}\right).$$

It is immediate to notice that the matrix has rank 1, therefore the geometric multiplicity of λ is 1. Therefore the Jordan form of F is

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -\frac{R}{2L} & 1 \\ 0 & -\frac{R}{2L} \end{bmatrix}.$$

This implies that the modes associated with λ are

$$e^{\lambda t} = e^{-\frac{R}{2L}t}, \qquad te^{\lambda t} = te^{-\frac{R}{2L}t}, \qquad t \in \mathbb{R}.$$

For $\Delta < 0$ one gets two complex conjugate roots,

$$\sigma \pm j\omega \doteq -\frac{R}{2L} \pm \frac{j}{2}\sqrt{\frac{4}{LC} - \frac{R^2}{L^2}},$$

and thus two sinusoidal modes, exponentially modulated

$$e^{\sigma t}\cos(\omega t), \qquad e^{\sigma t}\sin(\omega t), \qquad t \in \mathbb{R}.$$

As regards the internal and external stability of the system, it is immediate to notice that the characteristic polynomial $\Delta_F(s)$, previously computed, has all positive coefficients. Therefore, from the Descartes' rule of signs, it follows that all its roots lie in the open left half space. This ensures the asymptotic stability of the state space model, and hence its BIBO stability, independently of the (positive) values taken by the three parameters.

2.9 Exercises and Examples

Example 2.9.1 Let us consider the continuous time system whose unforced evolution is described by the following equation

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 4 & 0 \end{bmatrix} \mathbf{x}(t), \qquad t \in \mathbb{R}_+.$$

i) Determine the system elementary modes and specify their (convergent, limited, unlimited) behaviour;

ii) Determine the system unforced evolution starting from each of the following initial conditions:

- $\mathbf{x}(0) = [\,2 \quad -1 \quad 1\,]^T$;
- $\mathbf{x}(0) = [\,1 \quad 1 \quad 2\,]^T$;
- $\mathbf{x}(0) = [\,1 \quad 0 \quad 0\,]^T$.

Solution i) The explicit calculation of the characteristic polynomial leads to

$$\begin{aligned} \Delta_F(s) &= \det \begin{bmatrix} s-1 & -2 & 0 \\ 0 & s-1 & -1 \\ -2 & -4 & s \end{bmatrix} = s(s-1)^2 - 4 - 4(s-1) \\ &= s(s-1)^2 - 4s = s[(s-1)^2 - 4] = s[s^2 - 2s - 3] = s(s+1)(s-3). \end{aligned}$$

Since the three eigenvalues are distinct it is immediate to notice that the Jordan form of F is

$$J = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

to which three modes are associated: the bounded constant mode 1, the convergent mode e^{-t}, and the divergent mode e^{3t}.

ii) The vector $\mathbf{x}(0) = [\,2 \quad -1 \quad 1\,]^T$ is an eigenvector of the matrix F corresponding to the null eigenvalue. From the property

$$F\mathbf{v} = \lambda \mathbf{v} \qquad \Rightarrow \qquad e^{Ft}\mathbf{v} = e^{\lambda t}\mathbf{v}$$

it follows that the unforced state evolution corresponding to such an initial condition is

$$\mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}(0) = e^{0t}\mathbf{x}(0) = \mathbf{x}(0), \qquad t \geq 0.$$

Analogous considerations hold for the vector $\mathbf{x}(0) = [\,1 \quad 1 \quad 2\,]^T$ that is an eigenvector of F corresponding to the eigenvalue 3, therefore it holds that

$$\mathbf{x}_\ell(t) = e^{Ft}\mathbf{x}(0) = e^{3t}\mathbf{x}(0), \qquad t \geq 0.$$

For the third case, instead, we need to diagonalize the matrix F. To meet this goal we need to determine a non singular matrix T whose columns are three eigenvectors corresponding to the three eigenvalues $0, -1$ and 3, respectively. A suitable choice for T is:

$$T = \begin{bmatrix} 2 & 1 & 1 \\ -1 & -1 & 1 \\ 1 & 2 & 2 \end{bmatrix},$$

whose inverse is

$$T^{-1} = \begin{bmatrix} 2/3 & 0 & -1/3 \\ -1/2 & -1/2 & 1/2 \\ 1/6 & 1/2 & 1/6 \end{bmatrix}.$$

Therefore, it holds that

$$\begin{aligned} \mathbf{x}_\ell(t) &= e^{Ft}\mathbf{x}(0) = Te^{Jt}T^{-1}\mathbf{x}(0) \\ &= \begin{bmatrix} 2 & 1 & 1 \\ -1 & -1 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{3t} \end{bmatrix} \begin{bmatrix} 2/3 \\ -1/2 \\ 1/6 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} - \frac{e^{-t}}{2} + \frac{e^{3t}}{6} \\ -\frac{2}{3} + \frac{e^{-t}}{2} + \frac{e^{3t}}{6} \\ \frac{2}{3} - e^{-t} + \frac{e^{3t}}{3} \end{bmatrix}. \end{aligned}$$

♠

Example 2.9.2 Let us consider the continuous time system

$$\begin{aligned} \frac{d\mathbf{x}(t)}{dt} &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u(t), \\ y(t) &= [1 \quad 0]\,\mathbf{x}(t), \qquad\qquad t \in \mathbb{R}_+. \end{aligned}$$

i) Compute the Laplace transform $Y(s) = \mathcal{L}[y(t)]$ of the output signal $y(t)$, corresponding to the following conditions:
   - $\mathbf{x}(0) = [1 \quad 0]^T$ and $u(t) = e^{3t}, t \in \mathbb{R}_+$;
   - $\mathbf{x}(0) = [0 \quad 1]^T$ and $u(t) = 1 - t, t \in \mathbb{R}_+$.

ii) Determine, by making use of the Laplace transform, the input signal corresponding to the forced output signal $y(t) = te^t, t \in \mathbb{R}_+$.

Solution i) To determine the Laplace transform of the output signal, let us preliminary determine the Laplace transform of the input signal and of the system transfer matrix. In the first case it holds

$$U(s) = \frac{1}{s-3}$$

for the second one it holds

$$U(s) = \frac{1}{s} - \frac{1}{s^2} = \frac{s-1}{s^2}.$$

The system transfer matrix is

$$W(s) = H(sI_2 - F)^{-1}G = [\,1 \quad 0\,] \begin{bmatrix} s-1 & -2 \\ 0 & s-1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{s-3}{(s-1)^2}.$$

On the other hand, the unforced output evolution Laplace transform is

$$Y_\ell(s) = H(sI_2 - F)^{-1}\mathbf{x}(0) = [\,1 \quad 0\,] \begin{bmatrix} s-1 & -2 \\ 0 & s-1 \end{bmatrix}^{-1} \mathbf{x}(0) = \left[\, \tfrac{1}{s-1} \quad \tfrac{2}{(s-1)^2} \,\right] \mathbf{x}(0)$$

that leads in the first case to $Y_\ell(s) = \frac{1}{s-1}$ and in the second case to $Y_\ell(s) = \frac{2}{(s-1)^2}$. Therefore in the first case we get

$$Y(s) = \frac{1}{s-1} + \frac{1}{(s-1)^2}$$

that in the time domain corresponds to $y(t) = (1+t)e^t$, for $t \geq 0$, and in the second case we get

$$Y(s) = \frac{2}{(s-1)^2} + \frac{s-3}{(s-1)s^2} = \frac{2}{s} + \frac{3}{s^2} - \frac{2}{s-1} + \frac{2}{(s-1)^2}$$

that corresponds to

$$y(t) = [(2+3t) + 2(t-1)e^t],$$

for $t \geq 0$.

ii) It suffices observing that

$$U(s) = \frac{Y_f(s)}{W(s)} = \frac{1}{(s-1)^2}\frac{(s-1)^2}{s-3} = \frac{1}{s-3}$$

corresponds to $u(t) = e^{3t}, t \in \mathbb{R}_+$. ♠

Example 2.9.3 Let us consider the continuous time system described by the following equation:

$$\begin{aligned} \dot{x}(t) &= Fx(t) + Gu(t) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & a-2 & 0 \\ 0 & 1 & -2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} u(t) \\ y(t) &= Hx(t) = [\,1 \quad 0 \quad a-3\,]\, x(t), \end{aligned}$$

where a is a real parameter.

i) Determine, as a varies in $\mathbb{R}$, the Jordan form of the matrix F and study the asymptotic/simple and BIBO stability of the system.

ii) For $a = 0$, determine the forced state evolution when the input signal

$$u(t) = \begin{bmatrix} (1+2t) \cdot \delta_{-1}(t) \\ e^{-3t} \cdot \delta_{-1}(t) \end{bmatrix}$$

is applied.

SOLUTION i) It is immediate to verify that the matrix F has three eigenvalues, two of them located in -2 and one in $a-2$. Therefore, for $a=0$ the matrix has just one distinct eigenvalue of multiplicity 3, while for $a \neq 0$ it has two distinct eigenvalues, one of them of multiplicity 2, located in -2 and a simple one in $a-2$. Since the matrix is block diagonal, the Jordan form of F is the "direct sum" of the Jordan form of its miniblocks, namely for $a=0$ it is

$$J_{a=0} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

while for $a \neq 0$ it is (up to a rearrangement of the miniblocks)

$$J_{a\neq 0} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & a-2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

The system is asymptotically stable if and only if $a-2<0$, namely $a<2$, and it is simply stable if and only if $a \leq 2$. As regards the BIBO stability, the computation of the transfer function leads to

$$\begin{aligned} W(s) &= H(sI_3-F)^{-1}G = [1 \quad 0 \quad a-3] \begin{bmatrix} s+2 & 0 & 0 \\ 0 & s+2-a & 0 \\ 0 & -1 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \\ &= [1 \quad 0 \quad a-3] \begin{bmatrix} \frac{1}{s+2} & 0 & 0 \\ 0 & \frac{1}{s+2-a} & 0 \\ 0 & \frac{1}{(s+2)(s+2-a)} & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{s+2} & (a-3)\frac{s+3-a}{(s+2)(s+2-a)} \end{bmatrix} \end{aligned}$$

Obviously the system is BIBO stable for all the values of a for which the system is asymptotically stable. Moreover we notice that the first term of the transfer function is irrelevant for the study of the stability of the system (it is stable, and it remains such regardless of the value of a), therefore we only need to focus on the second term. We notice that for $a=3$ the second term of the transfer matrix is zero, therefore the system is BIBO stable even if it is not asymptotically stable. For any other value of a, instead, since the two factors $(s+3-a)$ and $(s+2-a)$ do not cancel out (and a possible cancellation of $(s+3-a)$ and $(s+2)$ is irrelevant since it would concern a "stable" factor), it follows that it is not possible to have BIBO stability when the system is not asymptotically stable. Therefore the system is BIBO stable for $a<2$ and $a=3$.

ii) By means of the Laplace transform we get

$$U(s) = \begin{bmatrix} \frac{s+2}{s^2} \\ \\ \frac{1}{s+3} \end{bmatrix}.$$

The Laplace transform of the forced state evolution is related to the Laplace transform of the input signal by the relation

$$\begin{aligned}
X_f(s) &= (sI_3 - F)^{-1}GU(s) = \begin{bmatrix} s+2 & 0 & 0 \\ 0 & s+2 & 0 \\ 0 & -1 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{s+2}{s^2} \\ \\ \frac{1}{s+3} \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{s+2} & 0 & 0 \\ 0 & \frac{1}{s+2} & 0 \\ 0 & \frac{1}{(s+2)^2} & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{s+2}{s^2} \\ \frac{1}{s+3} \end{bmatrix} = \begin{bmatrix} \frac{1}{s+2} & 0 \\ 0 & \frac{1}{s+2} \\ 0 & \frac{s+3}{(s+2)^2} \end{bmatrix} \begin{bmatrix} \frac{s+2}{s^2} \\ \\ \frac{1}{s+3} \end{bmatrix}. \\
&= \begin{bmatrix} \frac{1}{s^2} \\ \\ \frac{1}{(s+2)(s+3)} \\ \\ \frac{1}{(s+2)^2} \end{bmatrix}.
\end{aligned}$$

By taking the inverse Laplace transform of each term, one gets

$$\begin{array}{lcccl}
\frac{1}{s^2} & & & \to & x_1(t) = t \cdot \delta_{-1}(t) \\
\frac{1}{(s+2)(s+3)} & = & \frac{1}{s+2} - \frac{1}{s+3} & \to & x_2(t) = (e^{-2t} - e^{-3t})\delta_{-1}(t) \\
\frac{1}{(s+2)^2} & & & \to & x_3(t) = t \cdot e^{-2t}\delta_{-1}(t).
\end{array}$$

Therefore

$$\mathbf{x}_f(t) = \begin{bmatrix} t \\ e^{-2t} - e^{-3t} \\ t \cdot e^{-2t} \end{bmatrix} \delta_{-1}(t).$$

♠

Exercise 2.9.4 Let us consider the continuous time system described by the following equation:

$$\frac{d\mathbf{x}(t)}{dt} = F\mathbf{x}(t) + Gu(t), \qquad t \in \mathbb{R}_+,$$

where

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} \qquad \text{e} \qquad G = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

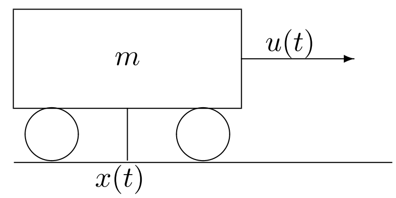
i) Determine the Jordan form of the matrix F, the elementary modes of the system and their (convergent, limited, unlimited) behaviour;

ii) determine the system state forced evolution when the input signal

$$u(t) = \delta(t) - e^{-t}\delta_{-1}(t)$$

is applied.

Exercise 2.9.5 Let us consider the continuous time system made up of a cart of mass m that moves under the effect of a traction force $u(t)$.



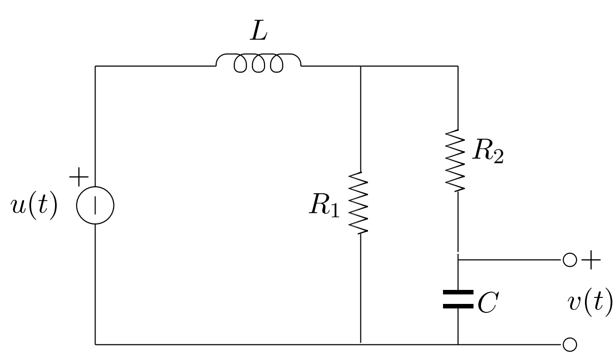
i) Determine a state space model that describes such a system, having the external force as input and the cart position as output [Indicate with k_f the viscous friction coefficient].

ii) Study the internal and BIBO stability of the system.

iii) Determine the system impulse response.

iv) Determine the values of the parameters m and k_f that guarantee that if the external traction force vanishes (i.e. becomes zero) at $t = 0$, the speed at $t = 10$ is not greater than the 10 % of the speed at time $t = 0$.

Exercise 2.9.6 Let us consider the following electric circuit.



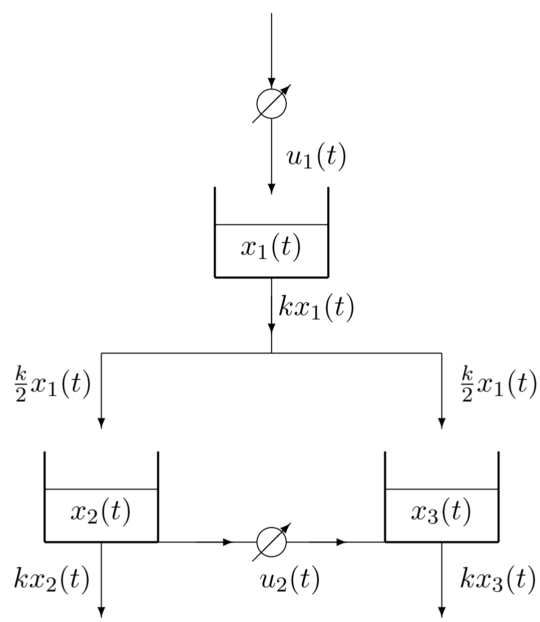
i) Determine a state space model that describes the circuit dynamics, having the voltage imposed by the voltage source as input and the voltage drop across the capacitor as output.

ii) Study the internal and BIBO stability of the system.

iii) Assuming $L = 10^{-3}\ H, C = 10^{-5}\ F$ and $R_1 = R_2 = R$, and assuming that starting from the time instant $t = 0$ the source imposes no voltage at the terminals, determine for which values of the parameter R the unforced output evolution is such that after 0.01 s the voltage $y(0.01)$ is always less than or equal to $e^{-10}y(0)$.

iv) For the values of L and C given before and by assuming $R_1 = 4\ \Omega$ and $R_2 \approx 0\ \Omega$, determine the impulse response of the system.

Exercise 2.9.7 Let us consider the hydraulic system in figure that consists of three tanks and two water pumps. The tanks are all identical and have an outlet capacity that is proportional to the volume of stored water. One of the two pumps is used to fill the tank on top (tank 1), while the other to transfer water from tank 2 to tank 3.



i) Design a state space model for the description of this system, having as inputs the water supply provided by the two pumps and as output the overall quantity of water in the system.

ii) Study the stability of the system.

iii) Determine the transfer function of the system.

Appendix A

The $\mathcal{Z}$-Transform and its Inverse Transform

A.1 $\mathcal{Z}$-Transform

If $v(k), k \in \mathbb{Z}$, is a real or complex sequence, then its **$\mathcal{Z}$-transform** is a function of the complex variable z defined by means of the expression

$$\mathcal{Z}[v(k)] \doteq \sum_{k=0}^{+\infty} v(k)\ z^{-k}.$$

The region of the complex plane in which the series converges is called **region of convergence** or RoC. It can be demonstrated that the region of convergence (if it is not empty) always includes the complementary set, in the complex plane, of a closed circle, centered in the origin, namely, the RoC includes a set of the type

$$\{z \in \mathbb{C} : |z| > r\}, \qquad r \in \mathbb{R}_+.$$

It can be proved (but we will not do it) that each $\mathcal{Z}$-transform is an analytic function in the interior of its RoC.

Since the $\mathcal{Z}$-transform takes into account only the samples of the sequence $v(k)$ corresponding to the time instants $k \in \mathbb{Z}_+$, from $\mathcal{Z}[v(k)]$ one can only reconstruct the portion of sequence corresponding to non negative time instants. As a consequence of the identity principle for powers series, it is possible to establish a one-to-one correspondence between causal sequences (by this meaning sequences that are zero for $k < 0$) and their $\mathcal{Z}$-transforms.

We want now to study the most interesting properties of the $\mathcal{Z}$-transform.

1) Linearity:

The $\mathcal{Z}$-transform is linear due to the linearity of the summation. Therefore if $V_i(z) = \mathcal{Z}[v_i(k)], i = 1, 2$, then, for any choice of the coefficients a_1 e a_2 in $\mathbb{C}$, one gets

$$\mathcal{Z}[a_1 v_1(k) + a_2 v_2(k)] = a_1 V_1(z) + a_2 V_2(z).$$

Moreover, if the RoC of $V_1(z)$ includes $|z| > r_1$ and the RoC of $V_2(z)$ includes $|z| > r_2$, then the RoC of $a_1V_1(z) + a_2V_2(z)$ includes $|z| > r$, where $r \doteq \max\{r_1, r_2\}$.

2) Multiplication by k:

If $v(k), k \in \mathbb{Z}$, admits $\mathcal{Z}$-transform, $V(z)$, for $|z| > r$, then the sequence $k \cdot v(k), k \in \mathbb{Z}$, admits $\mathcal{Z}$-transform as well, for $|z| > r$ and it is

$$\mathcal{Z}\left[k \cdot v(k)\right] = -z \cdot \frac{dV(z)}{dz}.$$

3) Time delay (or right translation):

If $v(k), k \in \mathbb{Z}$, admits $\mathcal{Z}$-transform, $V(z)$, then every delayed version of $v(k)$ admits $\mathcal{Z}$-transform. Specifically, if $d \in \mathbb{N}$ represents the delay, the delayed version by d steps of the sequence $v(k)$, has $\mathcal{Z}$-transform

$$\begin{aligned}\mathcal{Z}[v(k-d)] &= z^{-d}V(z) + \sum_{i=-d}^{-1} v(i)z^{-d-i} \\ &= z^{-d}V(z) + [v(-d) + v(-d+1)z^{-1} + \ldots + v(-1)z^{-d+1}].\end{aligned}$$

The right translation does not influence the RoC, in fact, the RoC of $V(z)$ coincides with the region of convergence of the new transform, except for the origin and at ∞. Notice that if the sequence $v(k)$ is zero for $k < 0$, the previous expression takes the form

$$\mathcal{Z}[v(k-d)] = z^{-d}V(z).$$

Notice that the one step delay corresponds, in terms of $\mathcal{Z}$-transform, to the product by z^{-1} (up to the initial condition $v(-1)$). For this reason it is common to represent the “one step delay” system with the symbol z^{-1}.

4) Time advance (or left translation):

If $v(k), k \in \mathbb{Z}$, admits $\mathcal{Z}$-transform, $V(z)$, then every time advanced version of $v(k)$ admits $\mathcal{Z}$-transform. Specifically, if $d \in \mathbb{N}$ represents the time advance, the advanced version by d steps of the sequence $v(k)$ has $\mathcal{Z}$-transform

$$\begin{aligned}\mathcal{Z}[v(k+d)] &= z^{d}V(z) - \sum_{i=0}^{d-1} v(i)z^{d-i} \\ &= z^{d}V(z) - [v(0)z^{d} + v(1)z^{d-1} + \ldots + v(d-1)z].\end{aligned}$$

The left translation does not influence the RoC, by this meaning that the RoC of $V(z)$ coincides with the region of convergence of the new transform, except for the origin and at ∞.

5) Multiplication by an exponential sequence:

If $v(k), k \in \mathbb{Z}$, admits $\mathcal{Z}$-transform $V(z)$ for $|z| > r$, then also the sequence $\lambda^k \cdot v(k), k \in \mathbb{Z}$, with λ an arbitrary complex number, admits $\mathcal{Z}$-transform

$$\mathcal{Z}[\lambda^k \cdot v(k)] = V\left(\frac{z}{\lambda}\right),$$

and such a transform converges for $|z| > r \cdot |\lambda|$.

6) Convolution:

If $v_1(k)$ e $v_2(k), k \in \mathbb{Z}$, are two sequences, zero for $k < 0$ that admit $\mathcal{Z}$-transform, indicated, with $V_1(z)$ and $V_2(z)$, respectively, then their convolution product $[v_1 * v_2](k), k \in \mathbb{Z}$, is well defined and admits $\mathcal{Z}$-transform

$$\mathcal{Z}\Big[[v_1 * v_2](k)\Big] = V_1(z)V_2(z).$$

The region of convergence includes the set $|z| > \max\{r_1, r_2\}$, where $|z| > r_i$ is the region of convergence of $V_i(z)$, $i = 1, 2$.

A.2 Examples of Common Z-Transforms

a) Kronecker delta (or discrete unitary impulse).

Let us consider first the discrete unitary impulse centered in 0. By definition of $\mathcal{Z}$-transform we get

$$\mathcal{Z}[\delta(k)] = \sum_{k=0}^{+\infty} \delta(k)\ z^{-k} = 1 \cdot z^0 = 1.$$

The RoC is, obviously, the overall complex plane. In case of a discrete impulse of the type $v(k) = \delta(k-i), i \in \mathbb{N}$, one gets

$$\mathcal{Z}[\delta(k-i)] = \sum_{k=0}^{+\infty} \delta(k-i)\ z^{-k} = 1 \cdot z^{-i} = z^{-i}.$$

The RoC in this case is $|z| > 0$. Notice that for $i < 0$, $\mathcal{Z}[\delta(k-i)] = 0$.

b) Discrete unitary step. The $\mathcal{Z}$-transform of the discrete time unitary step function $v(k) = \delta_{-1}(k)$ can be easily computed by a direct calculus:

$$\mathcal{Z}[\delta_{-1}(k)] = \sum_{k=0}^{+\infty} 1 \cdot z^{-k} = \frac{1}{1 - z^{-1}} = \frac{z}{z-1}, \qquad \text{for } |z^{-1}| < 1 \text{ namely } |z| > 1.$$

Notice, in fact, that for any fixed z whose modulus is grater than 1, the series defining the $\mathcal{Z}$-transform of $\delta_{-1}(k)$ is the geometric series with constant ratio z^{-1}.

c) Causal exponential sequence.

Consider $v(k) = \lambda^k\ \delta_{-1}(k)$, with λ complex number. The $\mathcal{Z}$-transform of v is obtained starting from the $\mathcal{Z}$-transform of the unitary step and by exploiting the property 5) on the multiplication by an exponential sequence. It is sufficient in fact to replace the variable z with the expression $\frac{z}{\lambda}$, thus obtaining

$$\mathcal{Z}[v(k)] = \sum_{k=0}^{+\infty} \lambda^k\ \delta_{-1}(k)\ z^{-k} = \frac{\frac{z}{\lambda}}{\frac{z}{\lambda} - 1} = \frac{z}{z-\lambda}.$$

In this case, the RoC is $|z| > |\lambda|$.

d) Causal sinusoidal sequence.

Set

$$v(k) = A\cos(\theta k + \phi)\ \delta_{-1}(k),$$

where A is a real positive integer number, while θ and ϕ are arbitrary real numbers. The $\mathcal{Z}$-transform of this signal is obtained by the $\mathcal{Z}$-transform of the causal exponential, derived at point c), by making usage of Euler's formula. So one gets

$$\begin{aligned}
\mathcal{Z}[v(k)] &= \mathcal{Z}\left[\frac{Ae^{j(\theta k+\phi)} + Ae^{-j(\theta k+\phi)}}{2}\ \delta_{-1}(k)\right] \\
&= \frac{Ae^{j\phi}}{2}\mathcal{Z}\left[(e^{j\theta})^k\ \delta_{-1}(k)\right] + \frac{Ae^{-j\phi}}{2}\mathcal{Z}\left[(e^{-j\theta})^k\ \delta_{-1}(k)\right] \\
&= \frac{Ae^{j\phi}}{2}\frac{z}{z-e^{j\theta}} + \frac{Ae^{-j\phi}}{2}\frac{z}{z-e^{-j\theta}} \\
&= \frac{A}{2}\frac{z[2\cos\phi\cdot z - 2(\cos\phi\cos\theta + \sin\phi\sin\theta)]}{z^2 - 2\cos\theta\cdot z + 1} \\
&= A\cdot\frac{z[\cos\phi\cdot z - \cos(\phi-\theta)]}{z^2 - 2\cos\theta\cdot z + 1}.
\end{aligned}$$

e) Causal exponential sequence multiplied by k.

Set $v(k) = k\ \lambda^k\ \delta_{-1}(k)$, with λ a complex number. The $\mathcal{Z}$-transform of this signal is obtained starting from the $\mathcal{Z}$-transform of the causal exponential, derived at point c), by applying the aforementioned property 2). So one gets

$$\begin{aligned}
\mathcal{Z}[k\lambda^k\delta_{-1}(k)] &= -z\cdot\frac{d}{dz}\left(\mathcal{Z}[\lambda^k\delta_{-1}(k)]\right) \\
&= -z\cdot\frac{d}{dz}\left(\frac{z}{z-\lambda}\right) = \frac{\lambda z}{(z-\lambda)^2}.
\end{aligned}$$

The region of convergence is still $|z| > |\lambda|$. From this property it is immediate to deduce that

$$\mathcal{Z}\left[\binom{k}{1}\lambda^{k-1}\right] = \frac{z}{(z-\lambda)^2}.$$

f) Exponential sequence multiplied by a binomial sequence.

Consider

$$v(k) = \binom{k}{\ell}\lambda^{k-\ell},$$

where ℓ is a negative integer and λ is a complex number. We want to demonstrate, by induction on ℓ and by exploiting the aforementioned properties, that for every $\ell \in \mathbb{Z}_+$ it holds

$$\mathcal{Z}\left[\binom{k}{\ell}\lambda^{k-\ell}\right] = \frac{z}{(z-\lambda)^{\ell+1}}.$$

The case $\ell = 0$ corresponds to example c), since for $\ell = 0$ we deal with a causal exponential sequence of the type λ^k. The case $\ell = 1$ has already been demonstrated in example e). Let us suppose now to have already proved the identity for a generic $\ell - 1 \geq 0$. We want now to prove that such property holds for ℓ, as well. In fact

$$\begin{aligned} v(k) &= \binom{k}{\ell}\lambda^{k-\ell} = \frac{k-\ell+1}{\ell}\cdot\frac{1}{\lambda}\cdot\binom{k}{\ell-1}\lambda^{k-(\ell-1)} \\ &= \frac{1}{\ell\lambda}\left[k\cdot\binom{k}{\ell-1}\lambda^{k-(\ell-1)} - (\ell-1)\cdot\binom{k}{\ell-1}\lambda^{k-(\ell-1)}\right] \end{aligned}$$

where it has been exploited the fact that

$$\binom{k}{\ell} = \frac{k-\ell+1}{\ell}\cdot\binom{k}{\ell-1}.$$

Therefore by applying the linearity, the inductive hypothesis and the property of the product by k, one gets:

$$\begin{aligned} \mathcal{Z}\left[\binom{k}{\ell}\lambda^{k-\ell}\right] &= \frac{1}{\ell\lambda}\left[-z\cdot\frac{d}{dz}\left(\frac{z}{(z-\lambda)^{\ell}}\right) - (\ell-1)\cdot\frac{z}{(z-\lambda)^{\ell}}\right] \\ &= \frac{1}{\ell\lambda}\left[-\frac{z\cdot(z-\lambda-\ell\, z)}{(z-\lambda)^{\ell+1}} - (\ell-1)\cdot\frac{z}{(z-\lambda)^{\ell}}\right] \\ &= \frac{z}{(z-\lambda)^{\ell+1}}. \end{aligned}$$

♠

A.3 The Inverse $\mathcal{Z}$-transform

We want now to focus our attention on the inverse $\mathbb{Z}$-transform of rational proper functions in the variable z and to do so we decompose the rational functions into "elementary components" whose inverse transform is known. It is worth saying that, since the $\mathcal{Z}$-transform uniquely identifies just the causal component of the sequence from which it has been obtained, we will adopt an inverse transform procedure such that the sequence $v(k), k \in \mathbb{Z}$, is zero for $k < 0$.

Let $V(z) \in \mathbb{C}(z)$ be a proper rational function, with distinct and non null (complex) poles $\lambda_1, \lambda_2, \ldots, \lambda_r$ with multiplicities $\mu_1, \mu_2 \ldots, \mu_r$ respectively and possibly a pole in the origin with multiplicity $\nu \geq 0$. It can be written in the following (irreducible) form:

$$V(z) = \frac{\bar{n}(z)}{z^\nu (z-\lambda_1)^{\mu_1}\ (z-\lambda_2)^{\mu_2}\ \ldots (z-\lambda_r)^{\mu_r}},$$

with $\bar{n}(z) \in \mathbb{C}[z]$ polynomial of degree at most $\nu + \sum_{i=1}^r \mu_i$. Notice that, if $V(z)$ has at least one pole $\lambda_i \neq 0$, the RoC of such a function includes the set $\{z \in \mathbb{C} : |z| > \max\{|\lambda_i|, i = 1, 2, \ldots, n\}\}$. If, instead, $V(z)$ has only poles in the origin, its RoC coincides with the overall complex plane except for the origin. Finally, in the specific case when $\nu = 0$ and $\sum_{i=1}^r \mu_i = 0$, $V(z)$ turns out to be a constant function, therefore its RoC coincides with $\mathbb{C}$.

By adopting a procedure to decompose a rational function into elementary fractions similar to the one adopted to obtain the inverse Laplace transform, we represent $V_1(z) \doteq V(z)/z$ (that is a strictly proper rational function) as the sum of elementary terms, namely in the form

$$V_1(z) = \frac{V(z)}{z} = \sum_{i=0}^{\nu} \frac{A_i}{z^{i+1}} + \sum_{i=1}^{r} \sum_{\ell=0}^{\mu_i - 1} \frac{C_{i,\ell}}{(z-\lambda_i)^{\ell+1}}.$$

The first summation does not appear if $\nu = 0$ and, if this is the case, $V(z)$ has a zero in the origin. The coefficients $C_{i,\ell}$ are determined in the same way as in the Laplace transform, by imposing the identity between the two representations of $V_1(z)$ multiplied by $z^{\nu+1}(z-\lambda_1)^{\mu_1}\ (z-\lambda_2)^{\mu_2}\ \cdots (z-\lambda_r)^{\mu_r}$.

So one gets

$$V(z) = A_0 + \frac{A_1}{z} + \ldots + \frac{A_\nu}{z^\nu} + \sum_{i=1}^{r} \sum_{\ell=0}^{\mu_i - 1} C_{i,\ell} \frac{z}{(z-\lambda_i)^{\ell+1}}.$$

In this expression we know the expression of the antitranform of each term, since

$$\frac{z}{(z-\lambda_i)^{\ell+1}} = \mathcal{Z}\left[\binom{k}{\ell} \lambda_i^{k-\ell}\right]$$

and

$$\frac{1}{z^i} = \mathcal{Z}[\delta(k-i)].$$

Therefore $V(z)$ is the $\mathcal{Z}$-transform of the following causal sequence:

$$v(k) = A_0\delta(k) + A_1\delta(k-1) + \ldots + A_\nu\delta(k-\nu) + \sum_{i=1}^{r}\sum_{\ell=0}^{\mu_i-1} C_{i,\ell}\binom{k}{\ell}\lambda_i^{k-\ell}.$$

To conclude, it is worth noticing that if the poles λ_i of $V(z)$ are simple, in order to determine the coefficients $C_i = C_{i,0}$ appearing in the previous decomposition it is sufficient to apply the strategy of the limit that has been already introduced for the Laplace inverse transform, namely

$$C_i = \lim_{z\to\lambda_i} V_1(z)(z-\lambda_i).$$

Example A.3.1 Let us consider the proper rational function

$$V(z) = \frac{z-1}{z(z+2)(z+1)}.$$

Let us compute, by means of the algorithm that we have just introduced, the causal sequence having $V(z)$ as $\mathcal{Z}$-transform. Let us define

$$V_1(z) \doteq \frac{V(z)}{z} = \frac{z-1}{z^2(z+2)(z+1)}.$$

The decomposition of $V_1(z)$ into partial fraction leads, after simple calculations, to:

$$V_1(z) = \frac{5/4}{z} - \frac{1/2}{z^2} + \frac{3/4}{z+2} - \frac{2}{z+1}.$$

Therefore

$$V(z) = 5/4 - \frac{1/2}{z} + \frac{3/4z}{z+2} - \frac{2z}{z+1},$$

is the $\mathcal{Z}$-transform of the sequence

$$v(k) = \frac{5}{4}\delta(k) - \frac{1}{2}\delta(k-1) + \frac{3}{4}(-2)^k\delta_{-1}(k) - 2(-1)^k\delta_{-1}(k).$$

♠

A.4 Exercises

Exercise A.4.1 Compute the $\mathcal{Z}$-transforms of the following signals:

(a) $v(t) = 1 + \delta(t) + \frac{1}{2^t}$;

(b) $v(t) = t \cdot \delta_{-1}(t) + 3^t\delta_{-1}(t+4)$;

(c) $v(t) = \delta(t+1) + \delta(t-1) + \binom{t}{2}$;

(d) $v(t) = \delta_{-1}(t) - \delta_{-1}(t-5) + \binom{t}{1}5^t$;

(e) $v(t) = \begin{cases} 1, & \text{for } t \in \mathbb{Z}_+, t \text{ even;} \\ 0, & \text{otherwise;} \end{cases}$

(f) $v(t) = \begin{cases} 2^t, & \text{for } t \in \mathbb{Z}_+, t \text{ odd;} \\ 0, & \text{otherwise;} \end{cases}$

(g) $v(t) = \begin{cases} 0, & \text{for } t = 0, 3; \\ 1, & \text{for } t = 1; \\ -1, & \text{for } t = 2; \\ 2, & \text{for } t \in \mathbb{Z}_+, t \geq 4; \end{cases}$

(h) $v(t) = \begin{cases} 1, & \text{for } t \in \mathbb{Z}_+, t \text{ even;} \\ -2, & \text{for } t \in \mathbb{Z}_+, t \text{ odd;} \end{cases}$

(i) $v(t) = \begin{cases} 1+t, & \text{for } t \in \mathbb{Z}_+, t \text{ even;} \\ 0, & \text{otherwise;} \end{cases}$

(l) $v(t) = \begin{cases} 2^t, & \text{for } t \in \mathbb{Z}_+, t \text{ multiple of 3;} \\ -1, & \text{for } t \in \mathbb{Z}_+, t \text{ congruous to 1 module 3;} \\ 0, & \text{otherwise.} \end{cases}$

Exercise A.4.2 Compute the inverse $\mathcal{Z}$-transform of the following proper rational functions:

(a) $V(z) = z^{-2} + \frac{1}{(z+4)(z+1)}$;

(b) $V(z) = \frac{z+1}{z(z-1)(z-2)}$;

(c) $V(z) = \frac{z^2}{z^2+10}$;

(d) $V(z) = \frac{2z}{z^2-\sqrt{3}z+1}$;

(e) $V(z) = \frac{z^2+1}{z(z+1)^2}$.